Chapter 1

Methodology

1.1 Introduction and Methodological Position

Chapter **??** established that automated lumbar MRI grading has moved beyond the stage at which novelty can be claimed from replacing one image backbone with another. Anatomical localisation, multi-sequence regions of interest, inter-level context and ordinal grading have all been demonstrated in recent work. The methodological question in this thesis is therefore not whether a sufficiently large network can classify lumbar MRI. It is whether the *structure of the prediction problem* can be represented more faithfully: which disease is being graded, at which level and side, which MRI sequence contains the relevant evidence, how neighbouring targets are related, and how the resulting model behaves when moved to a hospital whose scanners, protocols, population and reporting conventions were not represented during development.

The study is designed as a staged, controlled experimental programme around three methodological contributions and one translational validation programme. The first contribution is an anatomically aligned cross-sequence self-supervised representation in which positive relationships are defined by DICOM patient-space correspondence rather than by generic image similarity. The second is a disease-conditioned sequence router that learns how much each available MRI sequence contributes to each anatomical target and is trained explicitly to remain operational when one or more sequences are absent or technically present but degraded. Missingness and corruption are treated as distinct failure modes: an absent sequence is masked explicitly, whereas a corrupted sequence remains observable but is accompanied by quality evidence that the router may learn to distrust. The third is a heterogeneous disease–anatomy graph whose nodes correspond to level–condition–laterality targets and whose typed edges encode anatomical adjacency, within-level relationships and bilateral symmetry. The translational programme then separates development from deployment: the model is trained on a public multinational benchmark, frozen, evaluated zero-shot on an independent Rizgary Teaching Hospital cohort, and only afterwards adapted with controlled amounts of local labelled data.

Several components are deliberately treated as *enabling technology* rather than as doctoral contributions. Anatomical localisation, segmentation, 2.5D region construction, conventional convolutional or transformer encoders, class-imbalance handling, ordinal losses and calibration are all necessary to make the comparison scientifically fair, but none is advertised as new by itself. This distinction follows directly from the novelty boundary established in Chapter **??**: the thesis tests a new formulation of the structured task rather than presenting a collection of familiar modules as if their combination alone constituted originality.

The chapter is written as a protocol-grade methodology. Wherever an implementation choice has not yet been fixed by evidence, it is specified as a controlled comparison rather than invented retrospectively. Wherever an institutional fact remains unresolved—for example the final number of Rizgary studies surviving data reconciliation or the formal ethics reference— it is marked explicitly. This is intentional. A doctoral methodology should describe what is known, what is pre-specified and what will be decided by a transparent rule; it should not convert planning assumptions into apparently observed facts.

1.1.1 Research Questions

The five research questions derived in Chapter **??** are operationalised as follows.

RQ1 — Relational disease modelling. Does representing lumbar disease targets as a typed heterogeneous graph improve grading over independent target heads, an ordered five-level transformer and a homogeneous graph, and are any gains concentrated where single-target evidence is weakest?

RQ2 — Anatomically aligned self-supervision. Does cross-sequence pretraining based on DICOM-defined anatomical correspondence improve grading, label efficiency and transfer compared with ImageNet initialisation, generic medical pretraining and ordinary augmentation-based contrastive learning?

RQ3 — Disease-conditioned modality routing. Can one model learn target- and patient-dependent sequence weights while remaining robust to both missing and quality-degraded MRI sequences, without requiring a separately trained network for every combination of sagittal T1, sagittal T2/STIR and axial T2?

RQ4 — Cross-institutional transfer. What is the zero-shot degradation when a benchmark-trained model is applied to a previously unseen Middle Eastern clinical cohort, how much of that degradation is attributable to localisation versus grading/reference-standard shift, and how does performance recover as the amount of local annotation available for adaptation increases?

RQ5 — Clinically asymmetric error and uncertainty. Do ordinal and cost-sensitive objectives reduce clinically consequential distant errors, particularly Severe→Normal/Mild, and can calibrated uncertainty support selective prediction without overstating clinical safety?

The order is not arbitrary. RQ2 and RQ3 determine how evidence is represented; RQ1 determines how target predictions communicate; RQ5 determines how the ordered output and its uncertainty are represented; and RQ4 asks whether those choices survive the transition from a benchmark to real local clinical data.

1.1.2 Design Principles

Seven principles govern all experiments.

First, anatomy precedes grading. Every classifier and ablation receives the same anatomically localised input so that a comparison of representations is not confounded by one model being given a better crop than another. This follows the consistent evidence that localisation is a precondition of reliable per-level grading [**lu2018deepspine**, **pan2021automatically**, **batra2025mscan**].

Second, DICOM geometry is preserved. The primary scientific representation is reconstructed from DICOM patient-space coordinates. PNG conversion may be used for visual inspection but is not the basis of cross-sequence alignment, because it discards the spatial metadata required to determine which axial slices correspond to a sagittal disc level.

Third, patient identity defines independence. No images, slices, levels, sides or sequences from one patient may cross between training, validation and test partitions. The unit of statistical independence for uncertainty intervals and resampling is the patient, not the image or the disc level.

Fourth, the external cohort is not a tuning set. The primary model is developed without Rizgary images. Zero-shot performance is measured before any local tuning. Few-shot adaptation then uses a separately defined local adaptation pool while evaluation remains on a fixed local test set.

Fifth, every claimed contribution requires an ablation. A complete model score does not establish that anatomical self-supervision, routing or graph topology contributed anything. Each component is added and removed under a controlled experiment, and a negative result is reported as such.

Sixth, clinical interpretation is bounded by the reference standard. The model is trained to reproduce radiological grades whose inter-reader reliability is imperfect. Performance is therefore interpreted against human agreement and not as evidence that the network has discovered a more objective biological truth.

Seventh, each major dependency receives a diagnostic control. The primary end-to-end result is not replaced by manually corrected data, but a pre-specified subset is used to isolate failures that arise upstream of the claimed contribution. In particular, external grading is repeated with verified anatomical localisation, graph behaviour is stress-tested on isolated single-level disease, cross-sequence pretraining is tested for loss of sequence-specific information, and modality routing is evaluated under image corruption as well as complete absence. These controls are diagnostic rather than additional novelty claims.

1.2 Overall Research Design

1.2.1 Study Type

The project is a retrospective, multi-dataset medical-imaging methods study with three linked stages: public benchmark development, controlled methodological ablation, and independent cross-institutional transfer. It does not prospectively alter patient care and does not use model output to make treatment decisions. The Rizgary component is an external retrospective evaluation of imaging and radiology reports already acquired in routine clinical care.

The core development dataset is the RSNA Lumbar Degenerative Imaging Spine Classification benchmark (LumbarDISC), which contains multi-sequence MRI and expert per-level severity labels [**rsna2024rsna**, **richards2026the**]. SPIDER is used as an anatomical localisation/segmentation resource where its licence and task definition permit [**vandergraaf2024lumbar**]. The local Rizgary cohort is reserved for external transfer and domain-adaptation analysis rather than being mixed into the benchmark training data.

This separation is central to the inference the study is intended to support. A random split of one pooled dataset can estimate internal generalisation under that dataset's acquisition distribution. It cannot answer whether a model trained on international benchmark data transports to a different hospital. The latter requires the hospital to remain unseen during development.

1.2.2 Unit of Analysis

The primary unit of prediction on LumbarDISC is a *level–condition–laterality target*. At five disc levels (L1–L2, L2–L3, L3–L4, L4–L5 and L5–S1), the benchmark defines five targets: central canal stenosis, left and right neural foraminal narrowing, and left and right subarticular stenosis. Full benchmark scope therefore contains $5 \times 5 = 25$ targets per examination. Each target receives the ordered label Normal/Mild, Moderate or Severe [**richards2026the**].

The unit of *statistical independence*, however, remains the patient. Twenty-five outputs from the same examination are correlated measurements, not twenty-five independent observations. This distinction affects data splitting, confidence intervals, significance tests and error analysis throughout the chapter.

For the local Rizgary cohort, the target space is narrower. The reports contain disc morphology and central-canal descriptions but do not reproduce the entire RSNA schema reliably. The primary cross-institutional comparison is therefore restricted to the five central-canal targets, L1–L2 through L5–S1, for which a defensible overlap can be created. Bulge, protrusion and extrusion are analysed as a separate local morphology task rather than being relabelled as if they were RSNA stenosis grades. Foraminal and subarticular transfer are excluded unless the local reference standard is expanded by new expert grading.

1.2.3 Experimental Staging

The methodology is intentionally cumulative:

1. establish reproducible DICOM reconstruction, patient-level splits, localisation and an independent ROI baseline;
2. establish anatomically correct cross-sequence correspondence;
3. test disease-conditioned sequence routing and missing-modality training;
4. test anatomically aligned cross-sequence self-supervision;
5. test homogeneous and heterogeneous graph reasoning against independent target prediction;
6. test ordinal/cost-sensitive output heads and probability calibration;
7. freeze the selected public-data model and perform zero-shot external evaluation at Rizgary;
8. adapt with prespecified numbers of local cases and estimate the annotation–performance recovery curve.

The staging protects the thesis from a common failure mode of complex medical AI systems: if the final model underperforms, a monolithic experiment cannot reveal whether the failure arose from localisation, fusion, representation, graph reasoning, loss design or domain shift. Here each stage creates a measurable intermediate object and a direct baseline.

1.2.4 Pre-Specified Hypotheses

The principal hypotheses are directional but not expressed as arbitrary target accuracies.

H1 Anatomical cross-sequence self-supervision will improve label-efficient grading and cross-institutional robustness relative to generic pretraining, particularly when only a fraction of benchmark labels is used.

H2 Disease-conditioned routing with explicit availability masking, modality dropout and quality-aware gating will degrade less under missing- sequence and corruption experiments than fixed concatenation or ordinary cross-attention trained only on clean complete studies.

H3 A typed heterogeneous graph will outperform independent target heads and a homogeneous graph if the anatomical edge types carry useful relational information; performance under a shuffled-edge control should deteriorate if the topology itself matters.

H4 Zero-shot performance on Rizgary will be lower than internal benchmark performance because of scanner, protocol, population, localisation and reference-standard shift; the magnitude and pattern of that drop are treated as findings rather than failures of the project. A verified-ROI control is expected to separate the portion attributable to localisation from the residual grading-domain shift.

H5 Local adaptation will improve performance as annotation increases, but no predetermined percentage of "recovery" is required. The shape of the recovery curve and the location in the network at which limited adaptation is most effective are both treated as results.

H6 Cost-sensitive/ordinal objectives may reduce distant Severe→Normal/Mild errors even if they do not improve aggregate macro-F1; plain cross-entropy remains the mandatory baseline because previous lumbar work has shown that ordinal objectives do not automatically help [**niemeyer2021a**].

1.3 Data Sources

1.3.1 Public Development Cohort: LumbarDISC

The primary development data are drawn from the publicly released RSNA 2024 Lumbar Spine Degenerative Classification dataset. The published LumbarDISC resource comprises 2,697 patients and 8,593 MRI series collected across eight institutions in six countries [**richards2026the**]. The locally held labelled competition training subset contains approximately 1,975 studies and is used for core development. The distinction between the published cohort size and the available labelled subset is retained explicitly throughout the thesis rather than treating the two numbers as interchangeable.

Eligible benchmark examinations contain sagittal T2-like imaging, sagittal T1 and axial T2, with expert severity grading at the five lumbar levels [**rsna2024rsna**, **richards2026the**]. Coordinate/localiser annotations are also available for relevant disease locations. These properties make the dataset appropriate for all three methodological contributions: it is multi-sequence, level-resolved, multi-target, geographically heterogeneous and sufficiently large to separate representation learning from local external validation.

The benchmark serves three roles:

1. supervised development of per-target grading;
2. construction of anatomically corresponding cross-sequence training pairs;
3. internal/site-aware robustness analysis before the model is exposed to Rizgary.

All dataset versions, downloaded files, checksums where feasible and the final patient split lists will be archived in the reproducibility record. The exact number of studies remaining after local quality-control exclusions will be [**TO RECORD AFTER IMPLEMENTATION: final LumbarDISC development, validation and held-out counts**].

1.3.2 Public Anatomical Resource: SPIDER

SPIDER provides manually delineated lumbar anatomy on sagittal MRI from multiple hospitals, including vertebrae, intervertebral discs and spinal canal [**vandergraaf2024lumbar**]. It is used only for anatomical localisation, segmentation benchmarking or pretraining where permitted by its licence. It is not used to manufacture disease labels that are absent from the dataset.

The practical purpose is to avoid making localisation the doctoral novelty. Rather than spend the main experimental budget designing another segmentation network, the thesis will use or reproduce a strong established anatomical model and quantify its localisation error. If a SPIDER-pretrained model is used to initialise the localiser, the external disease-grading experiments remain separate because the disease targets originate in LumbarDISC.

1.3.3 Independent Local Cohort: Rizgary Teaching Hospital

The local cohort consists of lumbar MRI examinations and corresponding English radiology reports supplied by Rizgary Teaching Hospital. Current project inventory identifies 299 narrative reports and 294 candidate cases with matching multi-sequence imaging, while a wider raw DICOM audit contains additional studies. These counts are not treated as final until a one-to-one cohort reconciliation is complete.

The raw imaging inspected during protocol development confirms the sequence families required by the thesis—sagittal T1, sagittal T2 and axial T2—and preserves standard DICOM spatial metadata. A sample examination was acquired on a Siemens Avanto 1.5-T system; scanner/vendor distribution across the full local cohort will be reported after complete metadata extraction rather than inferred from one case.

The local cohort is valuable precisely because it is not curated to reproduce the public benchmark. Its scanner, acquisition protocol, prevalence, reporting language and label completeness differ from LumbarDISC. Those differences are the domain shift the transfer experiment is intended to measure, not nuisances to be eliminated by silently harmonising labels or discarding difficult cases.

1.3.4 Local Radiology Reports and Historical Spreadsheet

The narrative radiology reports are the primary textual source for the local reference standard. The existing spreadsheet is treated as a secondary historical transcription aid rather than as unquestioned ground truth. It contains patient-level variables and comma-separated level references, but it does not constitute a complete level-resolved RSNA-style matrix and has known transcription inconsistencies. The methodology therefore reconstructs the local reference matrix from source reports and uses the spreadsheet only for cross-checking.

The canonical local matrix will preserve, at minimum:

- pseudonymous case identifier;
- lumbar level;
- central-canal stenosis severity where explicitly stated;
- disc bulge, protrusion and extrusion as separate morphology fields;
- foraminal findings and side only where explicitly stated;
- nerve-root pressure/impingement where reported;
- an explicit missing/not-reported state distinct from a negative finding.

This last distinction is essential. Absence of a phrase in a narrative report is not automatically equivalent to “normal”. A field may be unreported because the radiologist considered it clinically irrelevant, because reporting practice varied, or because the report was not structured around that compartment. “Not reported” is therefore encoded separately and excluded from target- specific external evaluation unless a clinician has explicitly adjudicated the absence as normal.

1.3.5 Cohort Reconciliation

Before any local model evaluation, a case-flow table will reconcile the raw archive, reports and analysable imaging. For each candidate case the audit will record:

- whether a report exists;
- whether sagittal T1 exists and is usable;
- whether sagittal T2/STIR exists and is usable;
- whether axial T2 exists and is usable;
- whether DICOM geometry is internally consistent;
- whether a central-canal reference label can be derived;
- whether laterality-dependent labels are available;
- whether the case is duplicated or represents repeat imaging;
- the reason for any exclusion.

The final thesis will report the flow from raw studies to the fixed external test cohort, rather than presenting the current 299/294/raw-study counts without explanation. Repeat examinations from one patient, if identified, remain in the same partition to preserve patient independence.

1.4 Ethics, Governance and De-Identification

The Rizgary component uses retrospective clinical data. No MRI is acquired for the purposes of this study, no patient undergoes an additional intervention, and model output is not returned to the clinical record. Nevertheless, the data are sensitive medical information and are subject to institutional governance.

The final thesis will state the formal approval mechanism and reference: [**TO CONFIRM: hospital research committee/IRB approval number, date and responsible institution**]. It will also state the legal/ethical basis for retrospective use: [**TO CONFIRM: consent waiver, broad consent or other applicable basis**]. These items must be completed before the local study is described as ethically approved.

The raw local DICOM archive contains direct identifiers and therefore cannot be given to students or uploaded to external compute in its original state. De-identification is performed on a controlled copy before analysis. At minimum, the process removes direct patient identifiers, replaces case identifiers with study pseudonyms, strips or audits private tags, and handles dates according to the approved policy while preserving only the temporal information required for scientific analysis. The de-identification procedure must also check for burned-in identifiers where applicable. The re-identification key, if one must exist for hospital governance, is stored separately from the research repository and is not available to the model-training pipeline.

External/cloud processing of local images is prohibited unless explicitly allowed by hospital policy: [**TO CONFIRM: whether external cloud/GPU processing is permitted and under what conditions**]. Public benchmark data are handled according to their published licence and competition terms.

Only aggregate results are reported. Illustrative local images may be included in the thesis only after de-identification review and only if institutional permission permits publication. The local cohort is not assumed to be publicly shareable; any future data release would require separate approval and a more stringent disclosure review.

1.5 Reference Standard and Target Encoding

1.5.1 Benchmark Target Representation

For LumbarDISC, the target set is represented explicitly rather than collapsed into a generic three-class image label. Let the five lumbar levels be

$$\mathcal{L} = \{L1-L2, L2-L3, L3-L4, L4-L5, L5-S1\},$$

and let the five condition/laterality targets be

$$\mathcal{C} = \{\text{Canal}, \text{LF}, \text{RF}, \text{LSA}, \text{RSA}\},$$

where LF/RF denote left/right neural foraminal narrowing and LSA/RSA left/right subarticular stenosis. One examination therefore has the target set

$$\mathcal{T} = \mathcal{L} \times \mathcal{C}, \qquad |\mathcal{T}| = 25.$$

For target $t \in \mathcal{T}$ the observed grade is

$$y_t \in \{0, 1, 2\},$$

corresponding to Normal/Mild, Moderate and Severe. Encoding the targets in this form is not merely convenient notation. The pair (l, c) is retained throughout data loading, feature extraction, graph construction and evaluation, which prevents a severity label from losing the anatomical meaning that generated it.

Where the benchmark provides an absent or unusable annotation, a target mask

$m_t \in \{0, 1\}$ is carried alongside the grade. Masked targets do not contribute to the supervised loss. Missing labels are never imputed from neighbouring levels for the main experiments, because doing so would introduce exactly the relational assumption the thesis is intended to test.

1.5.2 Local Reference Matrix

The Rizgary reference standard is constructed at the same anatomical level but not forced into the same disease schema. A canonical row represents one case–level pair. The minimum schema is:

Field	**Type**	**Interpretation**
Case ID	categorical	pseudonymous patient/study identifier
Level	categorical	L1–L2 through L5–S1
Canal stenosis	ordinal/missing	explicit report-derived severity, if stated
Bulge	binary/missing	circumferential/generalised bulge stated at level
Protrusion	binary/missing	protrusion stated at level
Extrusion	binary/missing	extrusion stated at level
Foraminal finding	categorical/missing	narrowing/pressure effect if stated
Laterality	left/right/bilateral/NR	side only where stated
Nerve-root effect	categorical/missing	pressure, displacement or compression
Other morphology	text/coded	migration, dehydration, height loss and related findings

The local matrix keeps *not reported* (NR) separate from *explicitly absent*. This rule is especially important for foraminal and subarticular findings, because local reporting completeness differs from the structured RSNA annotation scheme. A target is eligible for external evaluation only when the reference standard provides a defensible class label.

1.5.3 Report Extraction and Human Verification

Automated report extraction, if available from the parallel clinical-NLP work, may accelerate matrix construction but does not define the reference standard. Every label that enters Selar's external grading analysis is manually checked against the original report text. This requirement applies to positive findings, explicitly negative/normal statements and the anatomical level to which the statement refers; an NLP output is never accepted as ground truth merely because its parser confidence is high. Label noise of this kind is well documented in medical image analysis, where labels derived from clinical text or from a single reader carry error rates high enough to change measured performance and to reward models that learn the annotation process rather than the disease [**karimi2020deep**]. The verification procedure below is designed so that residual noise is bounded and reported rather than absorbed silently into the reference standard.

Verification is organised in three levels, which deliberately distinguish *textual fidelity* from *clinical harmonisation*.

1. **Complete source-text audit.** Every externally evaluated case–level label is traced to the exact report phrase and its level and laterality. Multi-level expressions such as "L2–3 and L4–5, L5–S1 shows circumferential disc bulge" are expanded into separate level records without changing their meaning. Ambiguous phrases are coded as unresolved rather than forced into a class.

2. **Harmonised central-canal regrading.** The preferred confirmatory external reference standard is generated by two qualified readers who independently review the MRI for the fixed Rizgary test set and assign the same three-grade target used for the cross-institutional comparison (Normal/Mild, Moderate, Severe) at L1–L2 through L5–S1 using a common grading sheet. Readers are blinded to model predictions and, where feasible, to the original routine report during the first pass. Disagreements are adjudicated by a more senior reader or a pre-specified consensus procedure. Weighted κ [**cohen1968weighted**], exact agreement and the distribution of one-grade versus two-grade disagreements are reported.

   Adjudication is by explicit consensus rather than by statistical label fusion. Estimator-based fusion such as STAPLE [**warfield2004simultaneous**] is designed to recover both a reference standard and per-rater performance from a collection of raters, and is not adopted for a two-reader design: with two observations the performance parameters are weakly identified, and a disagreement is better resolved by a senior reader who can inspect the image than by an expectation-maximisation estimate over a pair of labels.

3. **Resource-limited reliability fallback.** If dual regrading of the entire fixed external test set is not feasible, a stratified subset is regraded before any model result is inspected. The size and sampling scheme are fixed in advance to cover lumbar levels and severity categories. The harmonised subset is then reported as the strongest external reference analysis, while report-derived labels on the remaining cases are explicitly labelled a secondary real-world analysis rather than treated as equivalent to the LumbarDISC reference process.

The historical spreadsheet is never used to override a source report or an adjudicated regrade. Where the report-derived label and harmonised reader label differ, both are retained with provenance so that reference-standard shift can be quantified rather than hidden. This separation is important because a cross-institutional performance drop may arise from image-domain shift, reference-standard shift, or both.

1.5.4 Schema Alignment Between LumbarDISC and Rizgary

Cross-institutional evaluation requires equality of *task*, not merely similar words. The mapping is therefore conservative.

LumbarDISC target	Rizgary evidence	Transfer status
Central canal stenosis	level-resolved mild/moderate/severe statements where present	Primary external target
Left/right foraminal narrowing	pressure/narrowing often incompletely lateralised	Exclude unless laterality and grade are adequate or freshly re-read
Left/right subarticular stenosis	not routinely represented in current reports	Exclude unless new expert annotation is created
Disc bulge/protrusion/extrusion	directly described locally but not an RSNA stenosis target	Separate local morphology task

No local disc-herniation morphology is converted into a stenosis grade by heuristic rule. Likewise, "pressure on exit foramina" is not automatically treated as a left/right foraminal severity label unless the report or expert review supplies the information required by that target. This prevents apparent external validation from being created by label re-interpretation.

1.6 Data Partitioning and Leakage Control

1.6.1 Patient-Level Independence

All splits are made at patient/study level before slice extraction or data augmentation. For a patient p, every sagittal series, every axial series, every disc level, every side and every augmented crop belongs to one and only one partition. A programmatic assertion is run before training to confirm that the intersection of patient identifiers across partitions is empty.

This discipline is not a formality. A systematic review of 294 papers across seventeen fields found leakage of exactly this kind—the same subject present on both sides of a split—to be among the most common causes of irreproducible machine-learning results in applied science [**kapoor2023leakage**]. The risk is acute in this study because a single patient contributes many slices, several sequences and up to twenty-five targets, so an image-level split would place near-identical data in training and test while appearing well formed.

The split record is version-controlled as a list of pseudonymous IDs. Data loaders consume those fixed lists rather than performing a new random split each time the training script is executed.

1.6.2 Public Development, Validation and Held-Out Evaluation

Where the available public dataset exposes institution identifiers, a site-aware evaluation is preferred because it tests a more meaningful shift than a random split. The exact public partitioning scheme will be chosen according to the metadata available in the held dataset and will be frozen before model comparison. At minimum there will be:

- a development/training partition;
- a validation partition for hyperparameter choice, early stopping and calibration;
- a held-out public evaluation partition not used for model selection.

If site identifiers permit leave-one-site-out or institution-held-out analysis, that analysis is reported in addition to the ordinary benchmark split. It is not a replacement for Rizgary, because the local hospital introduces a different reporting/reference-standard domain as well as a different imaging domain.

1.6.3 Fixed Local Test Set and Adaptation Pool

The local transfer study is protected against adaptation leakage by separating Rizgary into two parts *before* the public model is evaluated:

1. a **fixed local test set**, which is never used for adaptation, calibration or threshold selection; and
2. a **local adaptation pool**, from which the $N = 10, 25, 50, 100$ few-shot subsets are sampled.

The exact number allocated to each part will be determined after the audited class distribution is known. The split will be stratified by central-canal severity and lumbar level where feasible without producing clinically implausible or extremely small strata. The fixed local test set is used for the zero-shot baseline and for every subsequent adapted model, making the recovery curves directly comparable.

For each adaptation size N, several independent stratified subsets are sampled from the adaptation pool. Results are averaged across these draws with variance reported, because one set of ten cases can be unrepresentative by chance. The test set itself never changes across adaptation sizes.

1.6.4 Prevention of Information Leakage During Self-Supervision

Self-supervised learning does not remove the need for split discipline. A model pretrained on the held-out test patients, even without labels, has already seen their anatomy and acquisition characteristics. Therefore anatomical self-supervision uses only the development partition during internal experiments. The held-out public test set and fixed Rizgary test set remain unseen in both supervised and self-supervised training.

The same restriction applies to intensity normalisation parameters and learned harmonisation models: any transformation whose parameters are estimated from data is fitted on the training/development partition and then applied unchanged to validation/test data.

1.7 DICOM Reconstruction and Quality Control

1.7.1 Series Identification

Each study is first organised by DICOM `SeriesInstanceUID`. Candidate series are classified into sagittal T1, sagittal T2/STIR and axial T2 using a combination of DICOM metadata and image-orientation checks. Series descriptions are informative but not trusted alone because naming conventions differ across institutions. The classifier therefore considers fields such as `SeriesDescription`, `ProtocolName`, sequence parameters where available, and the orientation derived from `ImageOrientationPatient`.

Ambiguous or duplicated candidate series are flagged for manual review rather than selected by filename order. The selected series identifiers are stored in a manifest so that every experiment uses the same source series.

1.7.2 Patient-Space Geometry

For each DICOM slice, the image plane is reconstructed from `ImagePositionPatient`, `ImageOrientationPatient` and `PixelSpacing`. Let $\mathbf{s}$ be the patient-space location of the upper-left pixel, $\mathbf{r}$ and $\mathbf{c}$ the row and column direction cosines, and Δ_r, Δ_c the corresponding pixel spacings. The physical coordinate of pixel (i, j) is

$$\mathbf{x}(i, j) = \mathbf{s} + i\,\Delta_r\,\mathbf{r} + j\,\Delta_c\,\mathbf{c}. \tag{1.1}$$

The slice normal is

$$\mathbf{n} = \mathbf{r} \times \mathbf{c}. \tag{1.2}$$

Slices are ordered by projection onto $\mathbf{n}$ rather than by filename. Spacing between consecutive slice origins is inspected for discontinuities. This procedure preserves patient coordinates across sagittal and axial acquisitions, allowing a level localised in one plane to be projected into the other.

For a sagittal target point $\mathbf{q}$ and candidate axial slice k with origin $\mathbf{s}_k$ and normal $\mathbf{n}_k$, the perpendicular distance to that slice plane is

$$d_k = \left|\mathbf{n}_k^\top(\mathbf{q} - \mathbf{s}_k)\right|. \tag{1.3}$$

Distance alone is not treated as sufficient evidence of anatomical correspondence. Lumbar axial stacks are frequently prescribed obliquely so that the slice plane follows the local disc space. Let $\mathbf{n}_l^{\text{disc}}$ denote an estimated normal to the target disc plane, obtained from the local anatomical landmarks or endplate/disc geometry available from the localisation stage. The acquisition–disc angular discrepancy for slice k is

$$\theta_{k,l} = \arccos\!\left(\left|\mathbf{n}_k^\top\mathbf{n}_l^{\text{disc}}\right|\right). \tag{1.4}$$

A geometry relevance score combines distance and angular compatibility,

$$s_{k,l} = -\frac{d_k^2}{2\sigma_d^2} - \lambda_\theta\,\theta_{k,l}, \tag{1.5}$$

with corresponding normalised slice weights

$$w_{k,l} = \frac{\exp(s_{k,l})}{\sum_{j\in\mathcal{K}_l}\exp(s_{j,l})}. \tag{1.6}$$

The reference implementation therefore uses a small geometrically plausible candidate set $\mathcal{K}_l$ and supplies either the highest-scoring ordered slices or their weights to the downstream 2.5D aggregation module. This is preferred to forcing every clinical axial stack into a globally orthogonal grid, because resampling an intentionally disc-aligned acquisition can introduce through-plane interpolation and blur small neural structures. Canonical resampling may still be evaluated as a secondary robustness comparison, but it is not the default alignment strategy.

The per-level value θ_l (for example the median or maximum discrepancy within the selected stack) is retained as a quality-control variable. External error analysis tests whether unusually oblique or poorly aligned acquisitions are associated with localisation failure or grading error. The overall method remains an implementation of physical correspondence rather than a learned registration shortcut and follows the geometric principle used by prior lumbar pipelines [**pan2021automatically**, **batra2025mscan**].

1.7.3 Orientation Standardisation

Volumes are reoriented to a common anatomical convention for tensor processing while retaining the transform to original patient coordinates. Left/right must never be inferred after arbitrary array transposition; all laterality-dependent targets are checked against the DICOM orientation transform. This is particularly important for foraminal and subarticular labels, where a left/right inversion would create a scientifically silent but catastrophic error.

1.7.4 Voxel Spacing and Resampling

The acquisition spacing varies by institution, scanner and plane. Inputs are therefore resampled only after anatomical coordinates have been reconstructed. The target spacing is chosen from the training-set distribution so that it preserves structures relevant to grading without creating unnecessary memory cost. Sagittal and axial streams may use different in-plane resolutions because their native geometry and diagnostic roles differ.

Interpolation uses a continuous method (e.g. linear or higher-order interpolation) for MR intensities and nearest-neighbour interpolation for discrete masks or labels. Original spacing is retained in metadata for any measurement reported in millimetres. The exact target spacing will be [**TO RECORD AFTER IMPLEMENTATION: final sagittal and axial resampling spacings selected from the development cohort**].

1.7.5 Intensity Processing

MRI intensities are not quantitatively standardised across scanners. The preprocessing therefore removes gross scale differences without imposing a global histogram learned from test data. A default pipeline is:

1. suppress non-finite values and obvious padding/background values;
2. define a foreground mask from the body/spine region;
3. clip extreme foreground intensities using training-derived robust percentiles;
4. perform per-volume or per-ROI robust z-score normalisation;

5. optionally compare with bias-field correction or histogram standardisation as a robustness ablation.

Any learned harmonisation is fitted only on the development data. The external Rizgary zero-shot result is reported both with the baseline preprocessing and, if a separate harmonisation experiment is conducted, with that method clearly labelled as an intervention rather than folded into the zero-shot definition.

1.7.6 Quality-Control Flags

Automated QC records the following per series:

- slice count and ordering consistency;
- missing or duplicated instance positions;
- orientation consistency within series;
- axial-to-disc acquisition angulation discrepancy θ_l where target landmarks are available;
- in-plane spacing and slice spacing;
- field of view and lumbar coverage;
- intensity dynamic range;
- severe motion or truncation where detectable;
- presence of all sequences required by the particular experiment.

Cases failing geometry checks are not silently forced into the pipeline. They are reviewed, excluded with a recorded reason, or assigned to a robustness analysis if the failure itself represents a realistic missing-data condition. A final inclusion/exclusion table will report every such decision.

1.8 Anatomical Localisation and ROI Construction

1.8.1 Role of Localisation

Localisation exists to make the subsequent comparison fair. The thesis does not claim that identifying L1–L2 through L5–S1 is itself novel; Chapter 2 showed that this problem is mature and that strong systems already attain highly reliable level assignment [**pan2021automatically**]. The requirement here is that the localiser be accurate enough that representation and graph experiments are not dominated by wrong crops.

An established model is therefore preferred. Candidate implementations include a SPIDER-pretrained nnU-Net-style segmentation network [**isensee2021nnunet**, **vandergraaf2024lumb** a heatmap-regression detector, or another state-of-the-art anatomical parser available when implementation begins. The choice is made on validation localisation error and robustness, not on whether it is architecturally fashionable.

1.8.2 Outputs

The localiser produces, for each study:

- centres of L1–L2 through L5–S1 in patient coordinates;
- vertebral/disc/canal masks or landmarks where the selected model provides them;
- a confidence or quality flag for each detected level;
- the transform between voxel coordinates and DICOM patient space.

A failed or low-confidence localisation is retained as a distinct failure mode. It is not corrected manually for the primary automated pipeline without being reported, because doing so would inflate downstream grading performance by removing a real deployment error.

1.8.3 Localisation Evaluation

Where point annotations are available, centre localisation error is measured in millimetres:

$$e_l = \|\hat{\mathbf{q}}_l - \mathbf{q}_l\|_2. \tag{1.7}$$

Mean/median error and percentile distributions are reported by level. Percentage-of-correct-keypoint (PCK) or an equivalent tolerance-based measure is also reported at pre-specified physical thresholds. Where full masks are available, Dice and surface-distance metrics are reported. These results are separated from disease grading, consistent with the literature showing that detection/localisation and fine-grained grading generalise differently.

1.8.4 Localisation-Controlled External Analysis

Because localisation is an upstream dependency, a large external performance drop cannot automatically be attributed to the grading representation. A pre-specified stratified subset of the fixed Rizgary test set is therefore used for a diagnostic *verified-ROI control.* For this subset, qualified review confirms the lumbar level and corrects the target centre or crop only when the automated localisation is wrong. The primary external result remains the fully automated pipeline; manually verified ROIs are never substituted silently into that headline result.

The same frozen grading model is then evaluated twice on the control subset:

$$M_{\text{autoROI}} \qquad \text{and} \qquad M_{\text{verifiedROI}}, \tag{1.8}$$

and the localisation-attributable difference is reported as

$$\Delta M_{\text{loc}} = M_{\text{verifiedROI}} - M_{\text{autoROI}}. \tag{1.9}$$

A small ΔM_{loc} alongside a large public-to-Rizgary grading drop supports a representational/reference-standard domain-shift explanation. A large ΔM_{loc} instead shows that the enabling localiser is a material source of transport failure. The subset size and sampling scheme are fixed before the external grading results are inspected and are chosen to span lumbar levels, severity classes and, where possible, acquisition-quality conditions.

1.8.5 Disease-Specific 2.5D Regions of Interest

One rectangular crop is not assumed to be equally appropriate for every target. The crop definition is conditioned on anatomical compartment.

For central canal stenosis, the model receives sagittal T2/STIR context around the disc level together with corresponding axial T2 slices. For neural foraminal narrowing, the sagittal T1 stream receives greater parasagittal coverage and the ROI is side-aware, with sagittal T2 retained as complementary context. For subarticular stenosis, axial T2 supplies the dominant local evidence with sagittal context. These choices follow the clinical sequence– pathology correspondence established in Chapter 2 rather than being learned from arbitrary crop templates.

Each 2.5D stream contains a small ordered stack around the anatomically closest slice,

$$\mathcal{I}_l = \{I_{k-r}, \ldots, I_k, \ldots, I_{k+r}\},$$

where k is the level-centred slice and r is a small radius determined on the development set. A five-slice stack ($r = 2$) is the initial reference configuration because it captures through-plane context at modest memory cost; the final radius is reported after sensitivity testing rather than treated as a novel contribution.

Crops are defined in physical dimensions where possible, then resampled to the network input resolution. This avoids a fixed 100-pixel crop representing different anatomical widths on scanners with different pixel spacing.

1.8.6 ROI Quality Control

A validation subset is visually inspected using overlays of the detected level, crop boundary and corresponding axial slices. The inspection records:

- correct lumbar level;
- inclusion of the relevant canal/foramen/recess;
- correct left/right orientation;
- adequate axial coverage;
- crop truncation;
- correspondence failure caused by unusual axial angulation.

The purpose is not to manually curate the full dataset but to detect systematic geometry errors before they propagate into model training.

1.9 Baseline Representation and Sequence Encoders

1.9.1 Independent ROI Classifier

The mandatory baseline (E0) is an independent ROI classifier. Each target is graded from its anatomically localised input without communication with other targets. This represents the prevailing formulation criticised in Chapter **??** and provides the reference against which relational reasoning is measured.

The baseline uses a well-characterised image encoder such as ResNet [**he2016deep**], EfficientNet [**tan2019efficientnet**], Swin [**liu2021swin**], or another contemporary backbone selected before final experiments. The purpose is not to prove that one family is universally superior. A primary backbone is selected for the ablation ladder so that representation, routing and graph changes are measured under one stable feature extractor. A second backbone may be used in a model-agnosticity check for the strongest contribution.

1.9.2 Sequence-Specific Feature Extraction

Each available MRI sequence is processed by a sequence-specific encoder $E_m(\cdot)$,

$$\mathbf{f}_{p,l,m} = E_m(\mathcal{I}_{p,l,m}), \tag{1.10}$$

for patient p, level l and modality/sequence m. The encoders may share architecture while keeping separate normalisation statistics and weights, or may partially share weights if an explicit ablation shows that doing so is beneficial. Separate encoders are the default because T1, sagittal T2/STIR and axial T2 have different contrast and orientation statistics. In the standard taxonomy of multimodal learning, the problem posed by the MRI sequence stack is one of *alignment* and *fusion* rather than translation [**baltrusaitis2019multimodal**]: no sequence is to be generated from another, and the task is to place complementary views of one anatomy in a common frame and combine them. Naming the problem this way matters because it determines which prior work is a legitimate baseline.

To keep comparisons fair, the dimensionality of the output representation is held constant across fusion methods. Fixed concatenation, simple averaging and ordinary cross-attention all consume the same per-sequence features.

1.9.3 Controlled Backbone Selection

Backbone choice is selected on the validation partition before the central novelty experiments and then held fixed. The candidate comparison will use identical crops, augmentations, optimiser budget and patient split. If one backbone is selected because it gives the most stable validation performance rather than the absolute highest point estimate, that decision is recorded.

This design prevents the thesis from becoming a repeated architecture search. The main questions concern anatomical self-supervision, adaptive modality allocation and target-level relational reasoning. A backbone is a controlled feature extractor in those experiments.

1.9.4 Competitive Methodological Baselines

The internal E0–E8 ladder explains which proposed component contributes to a result, but it is not sufficient by itself to establish competitiveness with current lumbar MRI methods. Two additional reference implementations are therefore included where reproducibility permits.

1. **Multi-view cross-attention baseline.** A strong sagittal–axial fusion model following the design principle exemplified by M-SCAN is reproduced under the same patient split, ROI definitions and training budget used by the proposed system [**batra2025mscan**].

2. **Inter-level context baseline.** A model in which level-specific features communicate through an ordered Transformer, reflecting the recent anatomy-guided multi-level context literature, is trained under the same experimental conditions. This baseline tests whether the heterogeneous disease graph adds value beyond generic inter-level attention.

Published point estimates from these methods are reported only as contextual references when the original split or target definition differs. Claims of superiority require matched reimplementation or an otherwise genuinely comparable evaluation protocol; a higher number from a different split is not treated as a head-to-head result.

1.10 Core Contribution I: Anatomically Aligned Cross-Sequence Self-Supervision

1.10.1 Rationale

Conventional self-supervised vision learning defines related views by applying augmentations to one image, whether through contrastive objectives that contrast a positive against many negatives [**chen2020simple**, **he2020momentum**] or through non-contrastive self-distillation that dispenses with negatives entirely [**grill2020bootstrap**, **caron2021emerging**]. That is useful but does not exploit a distinctive property of multi-sequence MRI: sagittal T1, sagittal T2/STIR and axial T2 can look very different while representing the same underlying patient and spinal level. DICOM geometry supplies the correspondence directly.

The closest existing analogue is cross-modal contrastive alignment, in which paired observations in two modalities are pulled together in a shared embedding space [**radford2021learning**]. There the pairing is supplied by human-authored text; here it is supplied by patient-space geometry, which is both cheaper to obtain and, for the anatomical question at issue, more precise. The proposed objective is therefore a known learning mechanism applied to a pairing signal that the lumbar MRI literature has treated as a preprocessing convenience rather than as supervision.

The proposed objective therefore treats anatomical identity, rather than visual similarity, as the primary source of positive pairs. For patient p, level l, and two sequences m_a and m_b, embeddings $\mathbf{z}_{p,l,m_a}$ and $\mathbf{z}_{p,l,m_b}$ are encouraged to encode a shared anatomical representation even though acquisition plane and contrast may differ.

This idea is evaluated as a representation-learning hypothesis, not assumed to be beneficial. The literature already shows that generic and longitudinal self-supervision can reduce label dependence [**jamaludin2017selfsupervised**, **chen2020simple**], and that self-supervised pretraining improves both accuracy and label efficiency in medical imaging specifically, including a multi-instance variant that treats several images of the same pathology as positives [**azizi2021big**]; the experiment here asks whether explicitly anatomical cross-sequence correspondence is a stronger signal for lumbar grading.

1.10.2 Projection Head

Each sequence feature is mapped through a projection head P_m:

$$\mathbf{z}_{p,l,m} = \frac{P_m(\mathbf{f}_{p,l,m})}{\|P_m(\mathbf{f}_{p,l,m})\|_2}. \tag{1.11}$$

The projection space is used only to express the anatomical self-supervised constraint; the encoder features before projection are retained for routing and downstream grading. The projection head is discarded or decoupled after the pretraining stage unless an explicitly labelled joint-training ablation retains it. This separation reduces the risk that an objective designed to align anatomy across sequences will force the final disease representation itself to be sequence-invariant.

1.10.3 Positive-Pair Construction

The strongest positive relation is:

$$(p, l, m_a) \sim (p, l, m_b), \qquad m_a \neq m_b,$$

meaning the same patient and same anatomical level in two different sequences. Correspondence is established through the DICOM reconstruction and level localiser, not through nearest image appearance. To prevent artifact-degraded sequences from corrupting the contrastive representation, positive pairs are dynamically weighted by sequence quality scores $q_{p,m}$ (Section 1.7.6), excluding pairs where $q_{p,m} < \tau_{\text{quality}}$.

An optional softer relation may be tested for adjacent levels from the same patient, because neighbouring levels share biomechanical context. It is not included in the primary objective by default. Treating L3–L4 and L4–L5 as equivalent positives would risk erasing genuine level-specific pathology, so any adjacent-level term is isolated as a separate ablation.

Different patients at the same level are not defined as positives in the main objective, because common anatomical level does not imply common disease state. They may be used in a secondary supervised-contrastive variant once labels are available, but this is conceptually distinct from the proposed anatomical self-supervision.

1.10.4 Contrastive Objective

For anchor embedding $\mathbf{z}_i$ and its cross-sequence positive $\mathbf{z}_{i^+}$, the reference InfoNCE-style loss is

$$\mathcal{L}_{\text{ACSSL}}(i) = -\log \frac{\exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_{i^+})/\tau)}{\sum_{j \in \mathcal{B}\setminus\{i\}} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_j)/\tau)}, \tag{1.12}$$

where sim is cosine similarity, τ is a temperature parameter and $\mathcal{B}$ is the contrastive batch. When more than one anatomically valid positive exists, the numerator is extended to the set of positives rather than choosing one arbitrarily.

False negatives are a concern because two different patients may have very similar normal anatomy. The implementation will therefore compare ordinary InfoNCE with a variant that does not force all non-positive samples to be hard negatives, such as a multi-positive objective or a non-contrastive consistency objective of the kind that removes negative pairs altogether [**grill2020bootstrap**]. The latter is attractive here precisely because the false-negative problem is structural rather than incidental: normal lumbar anatomy is genuinely similar between patients, so a correct model should not be penalised for representing it similarly. The novelty claim is tied to the *anatomical pairing*, not to a particular brand of contrastive loss.

1.10.5 Label-Efficiency Experiment

The primary test of RQ2 uses fixed fractions of the labelled development set: 10%, 25%, 50% and 100%. For each fraction:

1. the encoder is initialised from the candidate pretraining strategy;
2. the same supervised grading head is fitted;
3. patient splits and target distribution are held fixed;
4. several random labelled subsets are evaluated where computationally feasible;
5. macro-F1, weighted κ, severe-class recall and calibration are compared.

The competing initialisations are:

- training from random initialisation;
- ImageNet initialisation;
- generic medical/volumetric pretraining where a suitable public model is available [**azizi2021big**];
- ordinary augmentation-based self-supervision;
- the proposed anatomically aligned cross-sequence self-supervision.

A successful result is not defined as beating every baseline at every label fraction. The useful scientific question is where anatomical pairing helps, where it does not, and whether any benefit is largest under label scarcity or domain shift.

1.10.6 Sequence-Specific Information Preservation Test

Anatomical alignment can become harmful if the encoder learns invariance by erasing contrast-specific information. The use of a projection head reduces this risk but does not eliminate it, because gradients from $\mathcal{L}_{\mathrm{ACSSL}}$ still update the sequence encoders. A dedicated preservation experiment therefore tests whether pretraining retains information that is diagnostically useful only in particular acquisitions.

For each major sequence, a single-sequence grading probe is trained from the same encoder with and without ACSSL pretraining while the probe architecture and labelled data are held fixed. Performance is compared separately on targets whose expected evidence is sequence-specific, including foraminal targets for sagittal T1 and canal/subarticular targets for axial T2. The experiment is not intended to prove a causal hierarchy among sequences. Its purpose is to detect a failure mode in which improved cross-sequence alignment is purchased by loss of sequence-unique diagnostic signal.

A representation is therefore considered beneficial only if any gain in label efficiency or transfer is not accompanied by a reproducible deterioration on these sequence-specific probes beyond the uncertainty of the comparison.

1.10.7 Transfer Test of the Representation

The public-data model used for zero-shot Rizgary evaluation is trained once with and once without anatomical self-supervision while the downstream architecture is held constant. If the anatomically pretrained model loses less performance externally despite comparable internal performance, that would support the claim that the representation is less tied to dataset-specific appearance. If both models degrade equally, the self-supervised contribution is limited to label efficiency rather than domain robustness.

1.11 Core Contribution II: Disease-Conditioned Adaptive Sequence Routing

1.11.1 Problem Formulation

A conventional multi-sequence model applies the same fusion rule to every target. That assumption is clinically implausible because the sequences are not equally informative for each anatomical compartment. Sagittal T1 is particularly useful for foraminal fat obliteration, sagittal T2/STIR for disc/canal context, and axial T2 for canal and lateral recess morphology. The router turns that known heterogeneity into a learnable, testable mechanism.

For patient p, target $t = (l, c)$ and available modalities $\mathcal{M}_p$, let $\mathbf{f}_{p,l,m}$ be the feature from modality m. The fused target feature is

$$\mathbf{u}_{p,t} = \sum_{m \in \mathcal{M}_p} g_{p,t,m}\, \mathbf{f}_{p,l,m}, \tag{1.13}$$

where the non-negative routing weights satisfy

$$\sum_{m \in \mathcal{M}_p} g_{p,t,m} = 1. \tag{1.14}$$

The gate is conditioned on target identity, normalised patient features and sequence-quality evidence. To avoid a high feature norm acting as an accidental proxy for importance, the image representation is first normalised,

$$\tilde{\mathbf{f}}_{p,l,m} = \mathrm{LayerNorm}(\mathbf{f}_{p,l,m}), \tag{1.15}$$

and the routing weights are

$$g_{p,t,m} = \mathrm{softmax}_{m \in \mathcal{M}_p} \left(G[\tilde{\mathbf{f}}_{p,l,m}, \mathbf{e}_c, \mathbf{e}_l, \mathbf{a}_p, \mathbf{q}_{p,m}] \right), \tag{1.16}$$

where $\mathbf{e}_c$ and $\mathbf{e}_l$ are learnable condition and level embeddings, $\mathbf{a}_p$ denotes optional acquisition/context metadata available without leaking the target label, and $\mathbf{q}_{p,m}$ is a compact quality representation derived from non-diagnostic acquisition/QC evidence such as motion score, truncation/coverage flags, spacing irregularity or other pre-specified image-quality indicators. The router remains patient-specific because image features are retained; the quality vector supplements rather than replaces them.

The gating mechanism itself has clear precedent. Squeeze-and-excitation recalibrates channel responses through a learned, content-dependent gate [**hu2020squeeze**], and sparsely-gated mixture-of-experts layers route each input to a subset of expert subnetworks through a trainable gating network [**shazeer2017outrageously**]. Neither is adopted unchanged. What is proposed here is that the gate be conditioned on *target identity*, so that one study is fused differently for a foraminal target than for a canal target at the same level. The novelty claim is the conditioning variable and its clinical justification, not the existence of a gate.

The initial implementation uses a lightweight MLP or target-conditioned attention gate. A more complex mixture-of-experts router is considered only if the simpler mechanism is insufficient. Should that route be taken, the load-balancing failure reported for sparsely-gated experts applies directly [**shazeer2017outrageously**]: without an explicit balancing term the gate can collapse onto one input and the remaining experts stop receiving gradient. The degenerate case is diagnostically important here, because a router that always selects sagittal T2 would be functionally identical to a fixed fusion rule while presenting as a learned one. Gate entropy and per-target weight distributions are therefore monitored during training and reported, and a collapsed router is recorded as a negative result rather than reported as successful routing.

1.11.2 Explicit Availability Mask

Missing sequences are represented by an availability mask $a_{p,m} \in \{0, 1\}$. Logits for unavailable modalities are masked before the softmax:

$$g_{p,t,m} = 0 \quad \text{if} \quad a_{p,m} = 0. \tag{1.17}$$

The remaining weights are renormalised over the modalities that actually exist. A missing sequence is therefore not represented by an all-zero image whose meaning the network must infer. Availability is explicit in the model.

This follows the hetero-modal principle, in which each modality is embedded independently and fusion is computed over whatever modalities are present, so that a single trained model accepts any subset without retraining [**havaei2016hemis**]. The masked-softmax renormalisation used here is a simpler realisation of that idea, chosen because the routing weights must remain interpretable as an allocation over available evidence, which moment-based fusion would obscure.

1.11.3 Modality Dropout

During training, complete studies are stochastically converted into incomplete ones by removing one or more sequence streams. Modality-wise dropout is an established route to test-time robustness against missing modalities that still permits cross-modal correlations to be learned during training [**neverova2016moddrop**]. It is adopted here because incomplete studies are the norm rather than the exception in the local cohort, so tolerance of absence is a deployment requirement and not merely a stress test. The dropout distribution includes:

- all three sequences;
- sagittal T1 + sagittal T2;
- sagittal T2 + axial T2;
- sagittal T1 + axial T2 where clinically meaningful;
- each single-sequence configuration as an edge case.

The sampling probabilities are tuned so that the model sees enough complete studies to learn complementary fusion while also receiving substantial exposure to incomplete inputs. The exact probabilities are [**TO RECORD AFTER IMPLEMENTATION: final modality- dropout schedule selected on validation data**].

No external test case is artificially repaired by copying another sequence into a missing slot. The model's purpose is to remain functional under genuine absence, not to conceal it.

1.11.4 Corrupted-Modality Robustness

A modality can be present yet clinically unreliable. This is distinct from missing-modality learning: an availability mask can suppress a sequence that is absent, but it cannot by itself protect the router from a motion-corrupted or truncated series that remains technically available.

During supervised routing experiments, controlled corruption is therefore applied to one sequence stream at a time using transformations that approximate plausible acquisition failures without manufacturing new pathology. Candidate corruptions include motion-like displacement/ghosting approximations, increased noise, bias-field/intensity distortion, partial field-of-view truncation and controlled slice loss. The exact corruption family and severity levels are pre-specified from the QC characteristics observed in the development data and are reported explicitly.

For each corruption, the analysis measures both prediction degradation and the change in routing allocation. If sequence m is corrupted, a robust router is expected, where clinically possible, to reduce $g_{p,t,m}$ relative to the clean input while redistributing weight to the remaining evidence. This expectation is evaluated rather than hard-coded. A model that preserves accuracy only by assigning high weight to a visibly corrupted sequence is treated as unstable even if aggregate performance appears acceptable.

Three controls are compared: routing without quality features, routing with $\mathbf{q}_{p,m}$ but no corruption training, and quality-aware routing exposed to controlled corruption during supervised training. The corruption experiment is a robustness test of RQ3 and not a separate novelty claim.

1.11.5 Routing Consistency

An optional consistency term encourages the prediction from an incomplete modality set to remain close to the complete-study prediction when the removed modality is non-essential:

$$\mathcal{L}_{\text{miss}} = D(p_\theta(y \mid x_{\text{full}}), p_\theta(y \mid x_{\text{subset}}))\,, \tag{1.18}$$

where D is a divergence such as KL divergence or Jensen–Shannon divergence. This term is not applied blindly: for targets known to depend strongly on a specific sequence, forcing invariance to its removal could suppress legitimate diagnostic information. The term is therefore ablated by target and compared with modality-dropout training alone.

1.11.6 Routing Baselines

RQ3 is evaluated against four progressively stronger controls:

1. fixed feature concatenation;
2. simple equal-weight averaging;
3. ordinary cross-attention over available sequence features;
4. disease-conditioned routing without modality dropout;
5. disease-conditioned routing with modality dropout.

For each method, encoder capacity, ROI inputs and training budget are matched as closely as possible. Missing-sequence robustness is evaluated on the same patients under the same artificial absence pattern.

1.11.7 Interpreting Routing Weights

The gate weights are model allocations, not causal estimates of sequence importance. A high axial T2 weight does not prove that axial T2 caused a correct decision. Clinical plausibility is assessed in two complementary ways.

First, aggregate routing weights are summarised by target. The expected pattern is that foraminal targets allocate more weight to sagittal T1 and subarticular/ canal targets more to axial T2, but this expectation is not built into the gate as a hard prior.

Second, controlled input ablation tests whether removing the sequence with high routing weight causes a greater performance loss than removing a low-weight sequence. Agreement between learned allocation and intervention strengthens the interpretation; disagreement is reported rather than rationalised post hoc.

1.12 Core Contribution III: Heterogeneous Disease–Anatomy Graph

1.12.1 Graph Construction

The proposed graph represents *prediction targets*, not pixels and not a 3D disc surface. At full LumbarDISC scope, the node set is

$$V = \{v_{l,c} : l \in \mathcal{L},\, c \in \mathcal{C}\}, \qquad |V| = 25. \tag{1.19}$$

Each node begins with the routed visual representation for that target together with embeddings that identify level and condition:

$$\mathbf{h}_{l,c}^{(0)} = [\mathbf{u}_{l,c} \| \mathbf{e}_l \| \mathbf{e}_c]. \tag{1.20}$$

The graph topology is defined anatomically before training. It is not learned from correlations in the test data.

The formalism is that of graph neural networks operating by neighbourhood feature propagation [**kipf2017semisupervised**]. The setting is strictly inductive: each patient presents a graph whose node features the model has never seen, so what is learned is an aggregation function rather than an embedding tied to one fixed graph [**hamilton2017inductive**]. Node set and edge topology are constant across patients; node features and evidence masks are not.

1.12.2 Node Evidence Availability Versus Label Availability

The graph distinguishes two forms of "missingness" that must not be conflated. A node may lack a ground-truth label in a particular cohort while still having valid image evidence, or it may lack usable visual evidence because a required sequence, field of view or localisation is missing. These cases are handled differently.

On LumbarDISC, a missing target label sets the supervised-loss mask $m_t = 0$ but does not automatically remove the node from the forward graph if the image inputs required to construct its representation are available. At inference, absence of a local Rizgary label likewise does not imply absence of the anatomical structure: the model is permitted to compute auxiliary foraminal or subarticular representations even when those outputs cannot be scored locally. Suppressing such nodes solely because their labels are unavailable would change the trained graph at external deployment and would confuse missing supervision with missing anatomy.

By contrast, when the visual evidence itself is invalid—for example the axial coverage does not include a target, localisation fails, or a required sequence is absent—an evidence mask $e_{p,t} \in \{0, 1\}$ is propagated to the graph. Edges carrying messages from nodes with $e_{p,t} = 0$ are masked before the attention softmax. This prevents genuinely unsupported "ghost" representations from contaminating neighbouring nodes while preserving information from unlabelled but observable anatomy.

1.12.3 Typed Edge Families

Three primary edge types are used.

Adjacent-level edges connect the same target type across neighbouring lumbar levels, for example

$$v_{L3-L4,\text{Canal}} \leftrightarrow v_{L4-L5,\text{Canal}}.$$

These encode the ordered chain of lumbar motion segments.

Same-level cross-condition edges connect anatomical compartments that coexist at one level, for example central canal to left/right foraminal and subarticular targets. The edge expresses that the targets share local morphology, not that one grade deterministically implies another.

Bilateral edges connect left and right counterparts of the same condition, for example

$$v_{L4-L5,\mathrm{LF}} \leftrightarrow v_{L4-L5,\mathrm{RF}}.$$

This represents bilateral anatomical symmetry without forcing equal grades.

Additional cross-type edges are added only if clinically justified in the pre-specified topology. A denser graph is not automatically a better graph; the study tests whether explicit relation types add value over homogeneous message passing. Distinguishing edge types by giving each relation its own transformation is the defining feature of relational graph convolution [**schlichtkrull2018modeling**]. The claim under test is therefore not that typed relations are a new idea, but that *these particular anatomical relations*—the motion-segment chain, same-level compartment coexistence and bilateral symmetry—carry information that undifferentiated message passing discards.

1.12.4 Relation-Aware Message Passing

For layer q, node i receives messages from neighbours $j \in \mathcal{N}(i)$. The formulation below combines learned attention over neighbours [**velickovic2018graph**] with relation-specific projections [**schlichtkrull2018modeling**], an arrangement close to the type-dependent attention used for heterogeneous graphs [**hu2020heterogeneous**]. A relation-aware attention formulation is

$$\mathbf{q}_i = W_Q \mathbf{h}_i^{(q)}, \tag{1.21}$$

$$\mathbf{k}_{j,r} = W_K^r \mathbf{h}_j^{(q)}, \tag{1.22}$$

$$\mathbf{v}_{j,r} = W_V^r \mathbf{h}_j^{(q)}, \tag{1.23}$$

with attention

$$\alpha_{ijr} = \mathrm{softmax}_{j,r}\left(\frac{\mathbf{q}_i^\top \mathbf{k}_{j,r}}{\sqrt{d}} + b_r\right), \tag{1.24}$$

and the neighbour message is

$$\mathbf{m}_i^{(q)} = \sum_{(j,r)\in\mathcal{N}(i)} \alpha_{ijr} \mathbf{v}_{j,r}, \tag{1.25}$$

with unavailable-evidence neighbours removed from the softmax by the mask $e_{p,j} = 0$. To reduce the risk that a focal abnormality is propagated indiscriminately into a normal adjacent target, local evidence and relational evidence are combined through a learnable residual gate,

$$\boldsymbol{\gamma}_i^{(q)} = \sigma\Big(W_\gamma[\mathbf{h}_i^{(q)} \| \mathbf{m}_i^{(q)}] + b_\gamma\Big)\,, \tag{1.26}$$

$$\mathbf{h}_i^{(q+1)} = \mathrm{LN}\Big(\mathbf{h}_i^{(q)} + \boldsymbol{\gamma}_i^{(q)} \odot \mathbf{m}_i^{(q)}\Big)\,, \tag{1.27}$$

followed by the usual feed-forward transformation. The gate allows a node with strong local visual evidence to retain its own state when neighbouring messages are inconsistent, while still permitting relational evidence to contribute when it is useful. The ungated residual formulation remains a mandatory ablation; otherwise any gain cannot be attributed specifically to the proposed control of neighbour influence. Relation-specific transformations W_K^r, W_V^r allow an adjacent-level neighbour to influence a node differently from its contralateral counterpart. Relation-specific parameters grow linearly with the number of edge families, a known source of overfitting in relational graph models. With three families over twenty-five nodes the risk is modest, but the basis-decomposition remedy [**schlichtkrull2018modeling**] is available should the pre-specified topology later be widened.

The exact graph transformer implementation may be replaced by a computationally simpler relation-aware GAT [**velickovic2018graph**] if validation shows no benefit from the more complex version. Where a transformer formulation is retained, lumbar level identity is made explicit to the attention mechanism through positional encodings [**dwivedi2021generalization**] or structural attention biases derived from graph distance [**ying2021transformers**], since the spine has a short, fixed path structure that such encodings represent naturally. The contribution is the explicit typed target graph and its controlled evaluation, not a requirement to use the largest graph architecture.

1.12.5 Graph Baselines

The graph contribution is tested against:

1. **Independent heads:** no message passing between targets.
2. **Ordered level transformer:** features communicate across the five lumbar levels but do not form 25 typed condition/laterality nodes.
3. **Homogeneous graph:** the same 25 nodes and adjacency pattern but one undifferentiated edge type [**kipf2017semisupervised**, **velickovic2018graph**].

4. **Heterogeneous graph:** typed adjacent-level, cross-condition and bilateral relations [**schlichtkrull2018modeling**, **hu2020heterogeneous**].

5. **Random/shuffled-edge control:** graph capacity retained while anatomical edges are permuted.

The shuffled-edge control is essential. If a graph with arbitrary topology performs as well as the anatomical graph, the study has shown that extra message-passing capacity helps, not that the encoded anatomy matters.

1.12.6 Edge-Family Ablation

The full graph is further decomposed:

- remove adjacent-level edges;
- remove same-level cross-condition edges;
- remove bilateral edges;
- retain only one family at a time.

Results are reported by target family. It is plausible, for example, that bilateral edges help lateral-compartment grading but add little to central-canal classification. A single average graph improvement would conceal this structure.

1.12.7 Isolated-Lesion Specificity Stress Test

A graph can improve average grading while still introducing a clinically undesirable failure mode: a severe focal lesion at one level may elevate the predicted severity of an adjacent level that is visually normal. This "information contagion" is evaluated directly rather than inferred from aggregate specificity.

The held-out public evaluation set is searched, without altering labels, for cases in which one target is Severe while the same target at one or both adjacent levels is Normal/Mild. For these naturally occurring isolated-lesion cases, the independent-head, homogeneous-graph, ungated heterogeneous-graph and gated heterogeneous-graph models are compared on the adjacent normal levels. The primary stress-test outcomes are the adjacent-level false-positive rate, the change in predicted Severe probability after graph refinement, and the fraction of cases in which the graph changes a correct local prediction into an incorrect higher grade.

A graph contribution is not considered clinically coherent if an aggregate gain is accompanied by a material, reproducible increase in adjacent-level false-positive contamination. This stress test is therefore part of the acceptance criterion for retaining relational message passing in the final integrated system. To safeguard against gate collapse caused by heavy class imbalance (where residual gating parameters $\boldsymbol{\gamma}$ lazily suppress relational propagation to zero), gate activation distributions $(\mu_\gamma, \sigma_\gamma)$ are tracked across training iterations, with an auxiliary entropy-maximization loss penalty applied if average gating falls below an operational threshold.

1.12.8 External Graph-Scope Sensitivity

Rizgary provides reliable evaluation labels primarily for central-canal stenosis, whereas the public graph is trained over all 25 targets. Absence of a local label for a foraminal or subarticular node does not make that node inherently invalid, but the possibility that auxiliary unvalidated nodes alter central-canal predictions is tested explicitly.

For the external central-canal analysis, three inference configurations are compared where implementation permits: (i) the full trained 25-node graph using all visually available node representations; (ii) a central-canal-only graph restricted to the five canal nodes using a separately trained or appropriately ablated model; and (iii) the full graph with only nodes lacking valid visual evidence masked. The comparison is interpreted as a sensitivity analysis of external graph scope, not as a post-hoc search for whichever configuration scores highest.

1.12.9 Graph Consistency and Uncertainty

Graph information is not used to hard-code clinically deterministic rules. Severe stenosis at L4–L5 does not require moderate stenosis at L3–L4, and the model is allowed to preserve isolated pathology. Relational message passing is therefore soft.

For uncertainty analysis, node disagreement can be measured between an independent-head prediction and the graph-refined prediction, or between neighbour-conditioned and local evidence. This disagreement is explored as an auxiliary signal for selective prediction, but it is not equated with calibrated clinical uncertainty unless validated against error.

1.13 Supporting Method: Ordinal and Cost-Sensitive Grading

1.13.1 Categorical Baseline

Standard three-class cross-entropy is retained as the mandatory reference:

$$\mathcal{L}_{\text{CE}} = -\sum_{t} m_t \sum_{k=0}^{2} \mathbb{I}(y_t = k) \log p_{t,k}. \tag{1.28}$$

Class weighting or balanced sampling is applied in separate controlled variants. This baseline is necessary because previous lumbar work found that explicitly ordinal objectives did not consistently outperform cross-entropy [**niemeyer2021a**].

1.13.2 Ordinal Threshold Formulation

An ordinal head represents a three-grade target as two ordered threshold questions:

$$P(y > 0 \mid \mathbf{h}), \qquad P(y > 1 \mid \mathbf{h}).$$

A CORAL/CORN-style implementation may be used provided the ordering constraint is respected. The exact ordinal formulation is selected before the final ablation and compared directly with categorical cross-entropy under identical features.

Performance is evaluated not only by exact class agreement but also by error distance:

$$d_t = |\hat{y}_t - y_t|. \tag{1.29}$$

The proportions with $d_t = 0$, $d_t = 1$ and $d_t = 2$ are reported separately.

1.13.3 Clinically Asymmetric Cost

A cost matrix C allows distant under-grading to receive a larger training penalty:

$$C = \begin{bmatrix} 0 & c_{01} & c_{02} \\ c_{10} & 0 & c_{12} \\ c_{20} & c_{21} & 0 \end{bmatrix}, \tag{1.30}$$

with the clinically motivated condition

$$c_{20} > c_{21},$$

so that Severe→Normal/Mild is more costly than Severe→Moderate. The exact numerical values are not invented by the model developer. They are either aligned with a defensible benchmark/clinical weighting or selected through a sensitivity analysis across a small pre-specified family of matrices. Specifically, baseline values for C are initialized to mirror the RSNA challenge evaluation weights ($1:2:4$ ratio for Normal/Mild, Moderate, and Severe grades), directly aligning the training penalty with the benchmark's clinical risk metric.

One implementation uses the expected cost under the predicted distribution:

$$\mathcal{L}_{\text{cost}} = \sum_t m_t \sum_{k=0}^{2} C_{y_t,k}\, p_{t,k}. \tag{1.31}$$

The cost term is evaluated alongside, not instead of, standard calibration and discrimination metrics. A model is not considered better merely because it optimises the chosen matrix.

1.13.4 Under-Grading Versus Over-Grading Trade-off

Cost sensitivity is not accepted as beneficial merely because it lowers the Severe→Normal/Mild rate. A sufficiently large penalty on under-grading can shift the decision boundary toward Severe and create an unacceptable false-positive burden. The cost-sensitive experiments therefore report both sides of the trade-off for each pre-specified cost matrix and each candidate value of λ_{cost}.

The minimum reporting set comprises Severe-class sensitivity, specificity, positive predictive value, the proportion of all targets predicted Severe, the Severe→Normal/Mild rate and the Normal/Mild→Severe rate, together with macro-F1, weighted κ and calibration. A cost-sensitive configuration is retained only if the reduction in clinically consequential under-grading is not explained by indiscriminate over-prediction of Severe.

Decision-curve analysis [**vickers2006decision**] is not a mandatory part of this imaging-grading study because the thesis does not observe a validated treatment or referral action threshold, and net benefit is defined only once such a threshold exists. If a clinically agreed binary action and threshold probability are established in a separately approved extension, decision-curve analysis may be reported there. Until then, the thesis reports the empirical sensitivity–false- positive trade-off without presenting radiological grades as treatment utility.

1.13.5 Class Imbalance

Because severe grades are uncommon in LumbarDISC, the study compares a small set of established strategies rather than stacking all of them simultaneously:

- unweighted cross-entropy;
- inverse-frequency or effective-number class weighting;
- weighted sampling;
- focal loss [**lin2017focal**];
- the cost-sensitive/ordinal variant above.

MixUp/CutMix are not the default for ordinal grading because mixed labels may have no clinical interpretation. Geometric augmentation is constrained so that left/right labels are updated consistently; horizontal flipping is prohibited for laterality-dependent targets unless labels and orientation are transformed together.

1.14 Supporting Method: Calibration, Uncertainty and Selective Prediction

1.14.1 Probability Calibration

Raw neural-network probabilities are not treated as calibrated confidence. Modern networks are systematically overconfident, and added capacity tends to worsen calibration even where it improves accuracy [**guo2017calibration**]. Temperature scaling is fitted on the validation set:

$$p_k^{(T)} = \frac{\exp(z_k/T)}{\sum_j \exp(z_j/T)}, \tag{1.32}$$

where $T > 0$ is selected without access to the test labels. Calibration is then evaluated with Expected Calibration Error, the Brier score [**brier1950verification**] and reliability diagrams.

1.14.2 Predictive Uncertainty

The primary feasible uncertainty methods are Monte Carlo dropout, which approximates Bayesian inference by retaining dropout at test time and so adds no training cost [**gal2016dropout**], and a small ensemble of independently initialised models, which is typically better calibrated and degrades more gracefully under distribution shift [**lakshminarayanan2017simple**]. The second property is the reason an ensemble is considered despite its cost, since the external Rizgary evaluation is itself a distribution shift and is where overconfidence would be most damaging. If S stochastic predictions are available, the predictive mean is

$$\bar{\mathbf{p}} = \frac{1}{S}\sum_{s=1}^{S} \mathbf{p}^{(s)}, \tag{1.33}$$

and uncertainty can be summarised by predictive entropy,

$$H(\bar{\mathbf{p}}) = -\sum_k \bar{p}_k \log \bar{p}_k, \tag{1.34}$$

together with between-member variance. The study reports whether uncertainty is higher on incorrect predictions, distant grade errors, low-agreement targets and external-domain cases.

1.14.3 Selective Prediction

A selective system abstains when uncertainty exceeds threshold δ:

$$U(x) > \delta \Rightarrow \text{refer/abstain.}$$

Rather than choosing one arbitrary threshold, the complete risk–coverage curve is reported: as increasingly uncertain cases are withheld, what fraction of the cohort remains covered and what error remains among the retained predictions? The area under, or summary points from, this curve can compare methods.

Conformal prediction offers a stronger guarantee than a heuristic threshold, converting any model into one that emits prediction sets with finite-sample coverage under exchangeability alone [**angelopoulos2023gentle**]. It is therefore reported for the internal held-out evaluation, where exchangeability is defensible. It is *not* claimed to hold on the Rizgary test set: that cohort is drawn from a different institution, scanner population and reporting process, which is precisely the violation of exchangeability under which the coverage guarantee fails. Where conformal sets are computed externally they are presented as a descriptive comparison against internally calibrated thresholds, with the broken assumption stated rather than left for a reader to notice.

The terminology "refer" describes model behaviour only. It does not claim a validated clinical workflow; a prospective reader study would be required to show that abstention improves patient care.

1.15 Composite Training Objective

The complete model may use the objective

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{grade}} + \lambda_{\text{ssl}}\mathcal{L}_{\text{ACSSL}} + \lambda_{\text{miss}}\mathcal{L}_{\text{miss}} + \lambda_{\text{cost}}\mathcal{L}_{\text{cost}} + \lambda_{\text{reg}}\mathcal{L}_{\text{reg}}, \tag{1.35}$$

where $\mathcal{L}_{\text{grade}}$ is categorical or ordinal supervised loss, $\mathcal{L}_{\text{ACSSL}}$ is the anatomical cross-sequence objective, $\mathcal{L}_{\text{miss}}$ is optional missing-modality consistency, $\mathcal{L}_{\text{cost}}$ is the optional clinical cost term and $\mathcal{L}_{\text{reg}}$ collects standard regularisation.

Graph message passing is normally part of the forward model rather than a separate loss. If an explicit graph-consistency regulariser is introduced, it receives its own coefficient and ablation.

Every non-zero λ is treated as a hyperparameter whose contribution must be demonstrated. A loss term is removed if it adds no reproducible value. The final thesis reports the selected values and the search range: [**TO RECORD AFTER IMPLEMENTATION: λ values, temperature, optimiser and regularisation settings**].

1.16 Optimisation and Training Protocol

1.16.1 Training Phases

Training is separated into conceptually distinct phases:

1. **Anatomical/localisation training or loading.** The localiser is trained/reproduced and then fixed for the grading experiments unless a joint-training ablation is explicitly conducted.

2. **Self-supervised pretraining.** Sequence encoders are trained with anatomical cross-sequence pairs using development patients only. The primary ACSSL phase uses studies for which the sequence pair required by the positive relation is present and passes QC. Modality dropout is disabled in this phase so that the pretraining objective is not changed by stochastic removal of its own positive pair. An incomplete-modality SSL variant, if investigated, is treated as a separate exploratory experiment with a dynamically defined set of valid positives.

3. **Supervised grading.** Encoders are fine-tuned with the routing and graph modules on LumbarDISC labels. Modality dropout and controlled corruption are introduced here, after anatomical pretraining, so that missing-modality robustness is learned without altering the definition of ACSSL.

4. **Calibration.** Temperature/uncertainty parameters are fitted on validation data after model selection.

5. **External evaluation.** The selected model is frozen and evaluated on the fixed Rizgary test set.

6. **Few-shot adaptation.** New model copies are adapted using only the local adaptation pool and re-evaluated on the same fixed local test set.

1.16.2 Optimiser and Schedules

AdamW [**loshchilov2019decoupled**] is the default optimiser for transformer/graph components; SGD or AdamW may be compared for CNN baselines if required. AdamW is preferred over Adam [**kingma2015adam**] because weight decay is decoupled from the adaptive step size, so regularisation strength can be tuned without confounding the effective learning rate—a distinction that matters when one schedule must serve encoder, router and graph modules of different scales. Learning rate is selected on the validation set within a pre-specified range, with warm-up and cosine decay [**loshchilov2017sgdr**] or a plateau scheduler used consistently across the main ablation. Early stopping is based on a prevalence-robust validation metric such as macro-F1 or weighted κ, not raw accuracy.

The final report records:

- optimiser and version;
- initial learning rate and scheduler;
- batch size / effective batch size;
- number of epochs and early-stopping patience;
- weight decay and dropout;
- input resolution and slice-stack radius;
- precision mode (FP32/BF16/FP16);
- random seeds;
- hardware and software versions;
- wall-clock and GPU time.

Mixed-precision training [**micikevicius2018mixed**] is used where it does not change results, because volumetric multi-sequence batches are memory-bound on the hardware available to this programme. At least one configuration is repeated in single precision to confirm that reduced precision has not altered a comparison on which a contribution claim rests.

No final hyperparameter value is fabricated in this protocol chapter; each is [**TO RECORD AFTER IMPLEMENTATION: executed training configuration and model-selection criterion**].

1.16.3 Augmentation

Training augmentation is designed to simulate plausible MRI variability without altering disease anatomy. Candidate transforms include modest intensity scaling, gamma/contrast change, bias-field simulation, noise, small translation/rotation and limited elastic deformation. Aggressive transformations that distort the canal or foraminal morphology beyond plausible acquisition variation are excluded.

Laterality-aware augmentation updates all side-specific labels. If a horizontal flip changes anatomical left/right, left and right target labels and graph node identity are swapped in the same transformation. Augmentation is disabled for validation and test data except in a separately labelled test-time-augmentation experiment.

1.16.4 Model Selection

Only validation data are used to choose architecture depth, routing hyperparameters, loss weights, early stopping and calibration. The held-out public test and fixed Rizgary test sets are not inspected for iterative model selection. If many exploratory experiments are performed, only the pre-specified final comparison is used for confirmatory statistical claims; exploratory results are labelled as such.

1.17 Experimental Ladder and Ablation Programme

The main experiment is not one comparison between "AMOG-Net" and a baseline. It is a ladder in which each additional assumption is isolated.

ID	Configuration	Question isolated
E0	Independent ROI classifier	How well can anatomically localised targets be graded without relational or adaptive fusion?
E1	+ DICOM-aligned multi-sequence ROIs	What is gained by correct cross-plane correspondence under fixed fusion?
E2	+ disease-conditioned sequence routing	Does target-specific routing outperform fixed fusion?
E3	+ modality dropout / quality-aware corruption training	Does explicit training for missing and degraded sequences improve robustness?
E4	+ anatomical cross-sequence SSL	Does anatomy-defined pretraining improve representation and label efficiency?
E5	+ homogeneous target graph	Does generic message passing help?
E6	+ typed heterogeneous disease–anatomy graph with gated residual update	Do clinically defined edge types add value without contaminating strong local evidence?
E7	+ ordinal/cost-sensitive/calibrated heads	Can clinically consequential errors and probability reliability be improved?
E8	Selected complete system + external transfer	Do the selected components survive institutional shift and adapt efficiently?

The ordering may be adjusted for engineering practicality, but the comparisons themselves remain. In particular, E6 must be compared with both E5 and a shuffled-edge control. Otherwise any improvement can be attributed to additional parameters rather than anatomical topology.

1.17.1 Contribution-Specific Ablations

Three ablation families correspond directly to the doctoral contributions.

Anatomical SSL ablation compares no SSL, generic image SSL and DICOM-aligned cross-sequence SSL under identical downstream training. It is repeated at reduced label fractions and accompanied by the sequence-specific preservation probes of Section 1.10.6.

Routing ablation compares fixed fusion, cross-attention, target- conditioned routing, target-conditioned routing with modality dropout, and quality-aware routing under controlled corruption. Performance is reported under complete, missing-sequence and corrupted-sequence input.

Graph ablation compares independent heads, an ordered five-level transformer, homogeneous graph, ungated heterogeneous graph, gated heterogeneous graph and random/shuffled edges. Edge-family removal further identifies whether adjacency, bilateral symmetry or same-level cross-condition interaction carries the signal, while isolated-lesion cases test whether any gain is purchased by adjacent-level contamination.

1.17.2 Negative Results

A component that fails to improve the pre-specified outcome is not silently removed from the research record. The thesis reports it, including confidence intervals and target-specific effects. This is particularly important for ordinal objectives and graph modelling, because both are theoretically attractive and therefore vulnerable to confirmation bias.

A component enters the final integrated system only if it adds a reproducible benefit, improves an outcome that the thesis has defined as clinically relevant, or yields a meaningful robustness/calibration advantage despite similar aggregate discrimination.

1.18 External Validation at Rizgary

1.18.1 Definition of Zero-Shot Transfer

Zero-shot transfer is defined strictly. The selected model, including encoder, router, graph, classifier and calibration strategy developed on public data, is frozen before any Rizgary test label is used. The local images are passed through the same preprocessing and localisation pipeline, with only non-learned DICOM reconstruction and pre-specified per-volume normalisation allowed.

No local fine-tuning, threshold optimisation or calibration is permitted in the zero-shot result. If a local intensity harmonisation method is fitted on Rizgary data, that constitutes a separate domain-adaptation experiment and is labelled accordingly.

The primary zero-shot analysis is restricted to the central-canal targets. Where the harmonised reader regrading described in Section 1.5.3 is available, that adjudicated three-grade matrix defines the confirmatory external reference standard. The manually audited routine-report matrix is retained as a secondary real-world reference analysis. The principal comparison is therefore not the full 25-target benchmark score versus a five-target local score; it is public held-out versus local external performance on the same central-canal grading task, with the reference-standard provenance stated explicitly.

1.18.2 Quantifying Domain Shift

Domain shift between institutions is a recognised and extensively catalogued failure mode in medical image analysis, with adaptation methods spanning supervised, semi-supervised and unsupervised settings [**guan2022domain**]. This study measures the shift before attempting to correct it, because its size and structure are themselves the finding sought by RQ4.

Performance degradation is reported as both absolute and relative change:

$$\Delta M = M_{\text{Rizgary}} - M_{\text{Public}}, \tag{1.36}$$

$$\Delta M_{\%} = 100\,\frac{M_{\text{Rizgary}} - M_{\text{Public}}}{M_{\text{Public}}}, \tag{1.37}$$

for each primary metric M. The thesis does not interpret a large negative ΔM as failed research; the size and structure of the drop are among the main outcomes of RQ4.

The shift is decomposed by:

- lumbar level;
- severity class;
- available sequence configuration;
- scanner/vendor/field strength where the local cohort contains enough variation;
- slice thickness/in-plane resolution;
- axial-to-disc angulation discrepancy and geometry quality;
- localisation success and verified-ROI correction status;
- report/reference-standard confidence and harmonised-reader status;
- image-quality/corruption indicators where available.

This analysis separates representational failure from data-quality failure where possible. For example, if grading declines but localisation remains stable, the pattern supports the detection-versus-grading asymmetry documented in Chapter 2. The interpretation is explicitly decomposed into at least three potential contributors: localisation shift, image/representation shift and reference-standard shift. The verified-ROI control estimates the first; the harmonised-reader analysis reduces the third; the residual difference is the closest available estimate of grading-domain transport failure, although the three sources cannot be made perfectly orthogonal in retrospective data.

1.18.3 Local Error Review

A blinded error audit is performed after the zero-shot metrics are locked. Incorrect local predictions are grouped into:

- wrong level/localisation;
- adjacent-grade disagreement;
- distant Severe↔Normal/Mild disagreement;
- apparent label/report ambiguity;
- missing or atypical sequence;
- image-quality/artifact issue;
- plausible model error despite adequate evidence.

Where reader time permits, a subset of model/report disagreements is re-read without exposing the model prediction. The purpose is not to alter the test labels opportunistically but to estimate how much apparent model error may reflect reference-standard variability.

1.19 Few-Shot Domain Adaptation

1.19.1 Adaptation Sizes

The adaptation pool is sampled at

$$N \in \{10, 25, 50, 100\}$$

patient-level labelled cases, subject to the final number of eligible local studies. Each N is sampled multiple times, stratified where feasible by central-canal severity. The same fixed local test set is used after every adaptation.

The experiment therefore estimates a function

$$M(N) = \text{test performance after adaptation with } N \text{ local cases}, \tag{1.38}$$

rather than a single “fine-tuned” number.

1.19.2 Adaptation Strategies

Few-shot adaptation is not treated as a single operation, because the dominant source of institutional shift may reside in the image representation, the modality router, the relational module or the final decision boundary. A pre-specified family of progressively less constrained strategies is therefore compared:

1. **No adaptation.** The frozen zero-shot baseline.
2. **Non-learned/simple harmonisation.** Intensity preprocessing is adjusted using only the adaptation pool where a defensible method exists.
3. **Head-only adaptation.** Only the final grading heads are updated.
4. **Router + head adaptation.** Sequence encoders and graph are frozen; routing and output layers are adapted.
5. **Graph + head adaptation.** Sequence encoders are frozen while relation-aware graph parameters and output heads are adapted.
6. **Late-encoder PEFT + head.** Low-rank [**hu2022lora**] or bottleneck-adapter [**houlsby2019parameterefficient**] parameters are inserted into a limited set of late encoder blocks while the remainder of the image representation stays fixed.
7. **Constrained multi-module PEFT.** A small parameter budget is distributed across late encoder, router and graph components according to a pre-specified configuration selected on development/domain-shift validation rather than the fixed Rizgary test set.
8. **Full fine-tuning.** All or most parameters are updated as a high-capacity reference only where N and compute make this defensible.

To evaluate the local morphology transfer task (disc bulge, protrusion, extrusion) under extreme sample constraints ($N \leq 25$), linear probing on frozen ACSSL representations is evaluated as a standalone baseline, comparing its transfer performance against full fine-tuning to directly test Hypothesis H1.

This design avoids assuming in advance that graph-only or encoder-only adaptation is correct. The trainable parameter count is reported for every strategy, and performance is interpreted jointly with that count. The exact adapter/LoRA implementation is supporting methodology taken directly from the parameter-efficient transfer literature [**houlsby2019parameterefficient**, **hu2022lora**], not a doctoral novelty; the scientific question is where a limited local adaptation budget is most effective and how that answer changes with N.

1.19.3 Annotation-Efficiency Summary

For each method, the recovery curve reports macro-F1, weighted κ, severe recall and calibration as a function of N. A useful summary is the area under the performance-versus-log-annotation curve, but raw points with confidence intervals remain primary because the curve may be non-linear.

No arbitrary threshold such as "95% recovery" defines success. If 100 local cases recover little performance, that is a meaningful negative finding about transportability. If 10 cases recover most of the loss, that is evidence for annotation-efficient adaptation.

1.19.4 Avoiding Adaptation Overfit

Few-shot training is particularly vulnerable to overfitting and hyperparameter selection bias, especially at $N = 10$. Repeatedly splitting ten cases into even smaller train/validation folds can itself create unstable model selection. The study therefore separates *method selection* from *few-shot estimation*.

- PEFT architecture, rank/adapter size, learning-rate range, maximum epoch budget and stopping rule are selected before the final $N = 10, 25, 50, 100$ recovery experiment, using public development/site-shift experiments or a separate local adaptation-development procedure that never touches the fixed Rizgary test set.
- Each N is repeated across several independent stratified adaptation samples; variance across these samples is reported because the identity of ten cases can dominate a point estimate.
- For the smallest N, the pre-specified stopping rule is used rather than inventing a two- or three-patient validation split. Where the adaptation pool is large enough, internal validation or cross-validation may be used for larger N as a secondary safeguard.
- The fixed external test set is never used to choose epoch, adapter rank, module placement, threshold, calibration or any other adaptation hyperparameter.
- Training-set loss and generalisation gap are reported for the tiny-N settings so that apparent recovery caused by memorisation is visible.
- PEFT is compared with full fine-tuning rather than assumed superior merely because fewer parameters are trainable.

1.20 Evaluation Metrics

No single metric is treated as sufficient because the task is imbalanced, ordinal and multi-target.

1.20.1 Macro F1 and Balanced Accuracy

For class k, precision and recall are

$$P_k = \frac{TP_k}{TP_k + FP_k}, \tag{1.39}$$

$$R_k = \frac{TP_k}{TP_k + FN_k}, \tag{1.40}$$

and

$$F1_k = 2\frac{P_k R_k}{P_k + R_k}. \tag{1.41}$$

Macro-F1 is the unweighted class mean,

$$F1_{\text{macro}} = \frac{1}{K}\sum_{k=1}^{K} F1_k, \tag{1.42}$$

so the severe class cannot be hidden by the Normal/Mild majority.

Balanced accuracy is

$$BA = \frac{1}{K}\sum_{k=1}^{K} R_k. \tag{1.43}$$

These are primary discrimination metrics.

1.20.2 Quadratic Weighted Kappa

Quadratic weighted κ [**cohen1968weighted**] is reported because the labels are ordered and the literature supplies human agreement values in κ. The weighting is not a technical detail but a clinical requirement: an unweighted κ would penalise a Normal/Mild-to-Moderate confusion exactly as heavily as a Severe-to-Normal/Mild one, collapsing the very distinction the grading task exists to preserve. For observed confusion matrix O and expected matrix E, with weight

$$w_{ij} = \frac{(i-j)^2}{(K-1)^2},$$

the statistic is

$$\kappa_w = 1 - \frac{\sum_{ij} w_{ij} O_{ij}}{\sum_{ij} w_{ij} E_{ij}}. \tag{1.44}$$

Exact agreement and within-one-grade agreement are also reported so that a high κ is not the only view of ordinal error.

1.20.3 Severe-Class Outcomes

The clinically decisive class is reported separately:

- Severe recall/sensitivity;
- Severe precision;
- Severe-versus-rest AUROC and AUPRC where appropriate;
- Severe→Normal/Mild error rate;
- Severe→Moderate error rate.

The separation distinguishes distant under-grading from adjacent boundary error.

1.20.4 AUROC and AUPRC

For three-class outcomes, AUROC is calculated one-versus-rest and macro-averaged. AUPRC is reported alongside AUROC for rare Severe classes because it is more sensitive to prevalence and false positives. Curves are constructed at patient- target level, while confidence intervals respect patient clustering.

1.20.5 Calibration

Calibration evaluation includes:

- Brier score [**brier1950verification**];
- Expected Calibration Error (ECE);
- classwise reliability diagrams;
- negative log likelihood where useful;
- risk–coverage curves under selective prediction.

Calibration is reported both internally and externally because a model can retain discrimination while becoming overconfident after domain shift.

1.20.6 Localisation Metrics

Localisation is evaluated separately using:

- mean, median and 95th-percentile Euclidean error in millimetres;
- PCK at physically meaningful tolerances;
- Dice coefficient for available masks;
- surface-distance metrics where mask quality supports them;
- complete localisation-failure rate per study.

1.20.7 Efficiency Metrics

For each model configuration the thesis reports:

- trainable and total parameter count;
- FLOPs/MACs where they can be estimated consistently;
- peak VRAM;
- training wall time and GPU hours;
- inference time per study and per target;
- storage size of weights.

Efficiency comparisons use the same hardware and batch/inference conditions. Timing from different machines is not compared as if it were an architectural property.

1.21 Statistical Analysis

1.21.1 Patient-Level Bootstrap Confidence Intervals

Confidence intervals for primary metrics are obtained by bootstrap resampling *patients*, not individual targets. On each replicate, a patient is sampled with replacement and all of that patient's target predictions enter together. This preserves within-study dependence.

The default is a sufficiently large number of replicates to stabilise the 2.5th and 97.5th percentiles; the final number is [**TO RECORD AFTER IMPLEMENTATION: bootstrap replicate count**]. Ninety-five per cent percentile or bias-corrected intervals are reported consistently.

1.21.2 Paired Model Comparisons

Ablation models are evaluated on the same patients, so comparisons are paired. For scalar patient-level losses or correctness indicators, a paired permutation test or Wilcoxon signed-rank test is used where distributional assumptions are not justified. AUCs are compared with an appropriate paired procedure such as DeLong's test [**delong1988comparing**] when its assumptions are satisfied. The paired form is required rather than optional here: ablation models are evaluated on the same patients, so their AUCs are correlated and an unpaired comparison would misstate the variance of the difference. Bootstrap differences are used for metrics without a simple analytic test.

The primary quantity is the paired difference

$$\Delta M = M_A - M_B$$

with a 95% confidence interval. Point estimates alone are not used to claim one component is superior.

1.21.3 Multiple Comparisons

The ablation programme contains many target-level analyses. A limited set of primary comparisons is therefore declared in advance:

1. anatomical SSL versus generic/self-supervised baseline;
2. routed+dropout fusion versus ordinary cross-attention;
3. heterogeneous graph versus independent heads and homogeneous graph;
4. zero-shot public versus local performance;
5. PEFT versus no adaptation across annotation sizes.

Target-by-target and subgroup analyses are secondary/exploratory. Where many formal hypotheses are tested, false-discovery-rate control is applied. The thesis distinguishes confirmatory comparisons from exploratory patterns instead of presenting every $p < 0.05$ as independent evidence.

1.21.4 Repeated Training Seeds

Deep-learning optimisation is stochastic. Final ablation comparisons are therefore repeated across multiple fixed seeds where computationally feasible. The thesis reports mean and standard deviation/interval across seeds in addition to the patient-level test uncertainty. A one-off training run is insufficient to attribute a small improvement to a method.

1.21.5 Subgroup Analysis

Subgroup analysis is conducted only where sample size permits and includes:

- lumbar level;
- severity;
- sex/age bands where metadata and governance allow;
- scanner/vendor/field strength;
- acquisition resolution or slice thickness;
- presence/absence of individual sequences.

To prevent reporting unstable point estimates on small strata, quantitative metrics (weighted κ, AUROC) are computed only for subgroups containing a predefined minimum sample size ($N \geq 30$); strata with $N < 30$ are reported exclusively via descriptive counts and raw confusion matrices.

Sparse cells are reported descriptively rather than subjected to unstable significance testing. The subgroup analysis is intended to expose failure modes, not to make causal claims about demographic biology.

1.22 Robustness Experiments

1.22.1 Missing-Sequence Matrix

The selected public-data models are evaluated under every supported sequence configuration. For three sequences, the core matrix includes complete input, three two-sequence combinations and three single-sequence configurations where the required anatomical target remains observable.

Performance is reported per target family rather than only in aggregate. A model can remain strong on central canal grading while collapsing on foraminal narrowing when sagittal T1 is removed; an overall mean would obscure the clinical meaning.

1.22.2 Geometry Perturbation

The reliance on DICOM coordinates creates a specific failure mode. Controlled perturbations therefore test sensitivity to small localisation/geometry errors: axial slice selection shifted by one or two slices, modest ROI-centre offsets, simulated spacing variation, and perturbation of the axial-to-disc angular compatibility term. Results are stratified by the recorded acquisition-angle discrepancy θ_l. The purpose is not to train on corrupted geometry by default, but to measure how brittle the system is to the imperfect localisation and oblique acquisition expected in practice.

1.22.3 Image-Quality and Acquisition Perturbation

Where feasible, controlled perturbations approximating scanner and acquisition failure are applied to the public held-out set: contrast scaling, noise, bias field, resolution degradation, motion-like corruption, slice loss and partial truncation. These do not replace real external validation but help identify which changes reproduce the pattern observed at Rizgary. For router experiments, the change in modality weight before and after corruption is reported alongside the change in prediction, so robustness is not inferred from accuracy alone.

1.22.4 Graph Perturbation

The graph is challenged by:

- random edge deletion;
- shuffled edge types;
- random rewiring with matched edge count;
- removal of one relation family.

A robust anatomical graph should outperform these controls without depending on one fragile edge. If random topology performs equivalently, the anatomical interpretation is rejected.

1.23 Interpretability and Failure Analysis

1.23.1 Local Visual Evidence

Grad-CAM or equivalent saliency may be used as a gross sanity check that the encoder responds within the relevant ROI, but saliency is not presented as a proof of reasoning. The stronger explanations in this thesis arise from structural quantities that are part of the model itself:

- which sequence received routing weight for a specific target;
- which graph relation contributed to target refinement;
- how uncertainty changed before and after relational reasoning;
- how prediction changed under controlled removal of a sequence or edge.

These are still model diagnostics, not causal explanations, but they are more directly linked to the proposed mechanisms than a heatmap alone.

1.23.2 Case-Level Audit Template

A fixed template is used for qualitative audit of representative correct and incorrect cases:

- ground-truth target and grade;
- predicted probability distribution and calibrated grade;
- localisation overlay;
- available MRI sequences;
- routing weights;
- dominant incoming graph relations/attention where extractable;
- uncertainty;
- nearest error category;
- whether the discrepancy is within one grade.

Cases are sampled from predefined categories rather than cherry-picked from the best-looking outputs.

1.24 Reproducibility and Software Engineering

1.24.1 Environment

The implementation will use a reproducible Python/PyTorch medical-imaging environment with DICOM tooling such as pydicom, SimpleITK and/or MONAI. The final thesis records exact versions of Python, PyTorch, CUDA, cuDNN, GPU hardware and all key libraries. Environment specification is stored in a lockfile or container recipe.

1.24.2 Experiment Configuration

Each run is configured from a machine-readable file containing:

- dataset manifest and split version;
- selected sequence IDs;
- preprocessing and resampling;

- ROI dimensions;
- encoder architecture;
- SSL/routing/graph settings;
- loss coefficients;
- optimiser and scheduler;
- seed;
- checkpoint and early-stopping rule.

The configuration is copied into the run directory with metrics and model weights so that a table in the thesis can be traced to an executable experiment.

1.24.3 Data Provenance

Raw data are never edited in place. A manifest records the path/identifier and processing status of every study. Derived crops can be regenerated from the manifest and DICOM source, and each derived sample retains its pseudonymous patient, level, condition, side, sequence and source-series identifier.

The local reference matrix is versioned independently of model output. If a label is corrected after clinical review, the change is documented and the affected experiments are rerun rather than silently updating test truth.

1.24.4 Code and Model Release

Code for the public-data methodology will be released where institutional and dataset licences permit. Pretrained weights are released if the training dataset terms allow redistribution. Local Rizgary images and report-derived individual-level data are not included unless a separate hospital approval authorises public release.

The repository will include:

- preprocessing and geometry reconstruction;
- split-generation/verification code;
- localisation/ROI pipeline;
- SSL, routing and graph modules;
- training and evaluation scripts;
- configuration files for primary experiments;
- instructions for reproducing public benchmark results.

1.25 Methodological Safeguards and Threats to Validity

1.25.1 Reference-Standard Shift

LumbarDISC uses structured expert grades; Rizgary originates from routine narrative reports. A performance drop can therefore reflect both imaging-domain shift and reference-standard shift. The thesis mitigates this by restricting transfer to defensible overlap, manually verifying every evaluated local label against source text and, for the strongest external analysis, constructing a harmonised central-canal reference through independent reader regrading with adjudication where feasible. Report-derived and harmonised-reader results are kept separate rather than averaged into one ambiguous reference standard.

1.25.2 Single-Centre Local External Validation

Rizgary is a genuinely independent institution but still only one local centre. External performance therefore demonstrates transport to *that* setting, not universal Middle Eastern generalisation. The thesis states this boundary explicitly. A future multi-hospital regional study is required before making a population-wide deployment claim.

1.25.3 Model Complexity

The full architecture contains several interacting components. Complexity can produce apparent gains merely through parameter count and tuning effort. The ablation ladder, matched backbone, homogeneous graph and shuffled-edge controls are specifically designed to separate mechanism from capacity.

1.25.4 Localisation Error

The graph and routing modules cannot recover evidence absent from a bad crop. Localisation is therefore measured and failures are reported separately. A verified-ROI control on a pre-specified Rizgary subset repeats grading after manual confirmation/correction of target localisation, allowing the external performance drop to be decomposed into an upstream localisation component and a residual grading/reference-standard component.

1.25.5 Graph Evidence and Unlabelled External Nodes

Rizgary cannot score every node used by the 25-target public graph. Missing local ground truth is not treated as missing anatomy, and nodes with valid visual evidence remain available to the trained graph. Truly unsupported nodes caused by absent sequences, failed localisation or inadequate coverage are masked. A full-graph versus canal-only sensitivity analysis tests whether unvalidated auxiliary nodes materially alter central-canal predictions.

1.25.6 Corrupted but Present Modalities

Availability masks solve absence but not poor quality. A motion-corrupted or truncated sequence can still enter the router and receive weight. Quality-aware gating, controlled corruption training and corruption-specific evaluation are therefore included, and routing weights are never interpreted as reliable when the model has not been tested under such degradations.

1.25.7 Class Imbalance

Severe grades are uncommon, so accuracy can be misleading and few-shot local subsets may contain few Severe examples. The primary metrics are macro-F1, balanced accuracy, weighted κ and per-class recall, with patient-level confidence intervals. Adaptation subsets are stratified where feasible and their severity counts are always reported.

1.25.8 Hyperparameter Multiplicity

A sufficiently large search can overfit the validation set even without touching test data. The study limits the number of tunable components in each stage, uses a fixed primary backbone for the core ablation, records the search space and freezes choices before external testing.

1.25.9 Clinical Utility

This methodology evaluates technical grading, calibration and portability. It does not demonstrate that radiologists become faster, safer or more accurate when using the model. Such a claim requires a prospective reader-assistance or impact study with independent clinical approval. The thesis therefore uses the language of "decision-support potential" rather than clinical benefit.

1.26 Mapping Research Questions to Evidence

RQ	Primary comparison	Primary outcomes	Failure interpretation
RQ1	independent/level transformer/homogeneous graph vs ungated and gated typed heterogeneous graph	macro-F1, κ, severe recall, edge ablations, isolated-lesion adjacent-level false positives	graph topology carries little useful signal or relational messages contaminate strong local evidence
RQ2	generic pretraining vs anatomical cross-sequence SSL at 10/25/50/100% labels	macro-F1, κ, label efficiency, external drop, sequence-specific preservation probes	anatomy-defined positives do not improve transferable representation or erase modality-specific signal
RQ3	fixed fusion/cross-attention vs routing, modality dropout and quality-aware routing	missing- and corrupted-sequence degradation, target-specific performance, routing-weight response, calibration	fixed fusion is sufficient or learned routing is unstable under absence/corruption
RQ4	public held-out vs fixed Rizgary test; automated vs verified ROIs; report-derived vs harmonised labels; module-wise PEFT vs no/full tuning	zero-shot Δ, localisation-controlled ΔM_{loc}, recovery curve, calibration shift	transport failure is dominated by localisation, reference-standard shift or representation shift; adaptation may remain inefficient
RQ5	CE vs ordinal/cost-aware objectives; calibrated vs uncalibrated prediction	Severe under-grading and over-grading rates, Severe PPV/specificity, grade distance, Brier/ECE, risk–coverage	lower under-grading is offset by unacceptable Severe false positives or ordinal/cost formulation adds no advantage

This table defines what evidence can answer each question. It also makes the thesis robust to partial negative results: RQ1 can be answered if the graph does not help, and RQ2 can be answered if generic pretraining is equal or superior. The contribution is the controlled experiment and the resulting knowledge, not a requirement that every proposed mechanism win.

1.27 Planned Reporting Standard

The final manuscript/thesis reporting will follow the current applicable medical-AI and diagnostic/prediction reporting checklists at the time of submission. Four apply to different parts of this work and are not interchangeable. TRIPOD+AI [**collins2024tripodai**] governs the development and validation of the prediction model. CLAIM [**tejani2024checklist**] governs the medical-imaging specifics, including acquisition, ground-truth definition and model description. STARD [**bossuyt2015stard**] governs any analysis reported as diagnostic accuracy against a reference standard, including the harmonised reader study. STROBE [**vonelm2007strengthening**] governs the description of the retrospective Rizgary cohort itself. At minimum the report will include:

- complete cohort flow diagrams and inclusion/exclusion reasons;
- reference-standard source, annotator qualifications and agreement where available;
- exact patient-level split logic and IDs reproducible within the approved environment;
- all primary metrics with confidence intervals;
- per-class Severe performance and ordinal error distances;
- calibration and external-domain performance;
- ablations for every claimed novel component;
- repeated-seed variability;
- compute/hardware cost;
- explicit limitations and negative results;
- a statement of what was not tested clinically.

The reporting checklist itself will be versioned with the manuscript because CLAIM/STARD/TRIPOD guidance may be updated during the candidature.

1.28 Chapter Summary

This chapter translates the research gap of Chapter **??** into a controlled experimental design. The study begins from a structured definition of lumbar disease: twenty-five level–condition–laterality targets on the public benchmark rather than a generic image-level class. DICOM patient-space geometry is preserved so that sagittal and axial evidence can be aligned physically; localisation and disease-specific 2.5D ROIs are treated as shared enabling technology; and all data partitions are defined at patient level.

Three methodological contributions are then tested independently. Anatomically aligned cross-sequence self-supervision asks whether the same spinal level viewed under different MRI sequences provides a stronger representation-learning signal than generic augmentation. Disease-conditioned routing asks whether one model can allocate different sequence evidence to different targets while remaining functional when modalities are absent or degraded, with quality-aware gating tested under controlled corruption. The heterogeneous disease–anatomy graph asks whether explicit typed relations among adjacent levels, co-occurring compartments and bilateral targets improve grading beyond independent heads, an inter-level transformer or a homogeneous graph without inducing adjacent-level information contamination. Anatomical SSL is additionally tested for preservation of sequence-specific diagnostic information. Ordinal, cost-sensitive and uncertainty-aware outputs remain supporting methods and are retained only if their contribution is measurable and does not simply exchange severe under-grading for indiscriminate over-grading.

The translational design is deliberately separated from model development. Rizgary is not mixed into public training. A fixed local test set establishes zero-shot domain shift, while a separate adaptation pool supplies controlled few-shot subsets. External interpretation is strengthened by a verified-ROI control that isolates localisation error and by a harmonised central-canal reader reference where feasible, while the report-derived matrix is retained as a distinct real-world reference analysis. Few-shot adaptation compares where a restricted update budget is applied—head, router, graph or late encoder—rather than assuming one PEFT location is universally optimal. The primary local comparison remains restricted to central-canal stenosis; local herniation morphology is analysed separately rather than being forced into the RSNA schema.

Finally, the methodology defines success as an answer to the research questions, not as attainment of a predetermined accuracy. A graph that adds nothing, an ordinal loss that fails to beat cross-entropy, or a large cross-institutional performance drop are all informative if demonstrated under the same controlled protocol. This principle is what turns the architecture from an engineering assembly into a doctoral experiment: every claimed mechanism is falsifiable, every comparison is tied to a research question, and the external cohort is used to test portability rather than to decorate an internal result.